Feature Extraction

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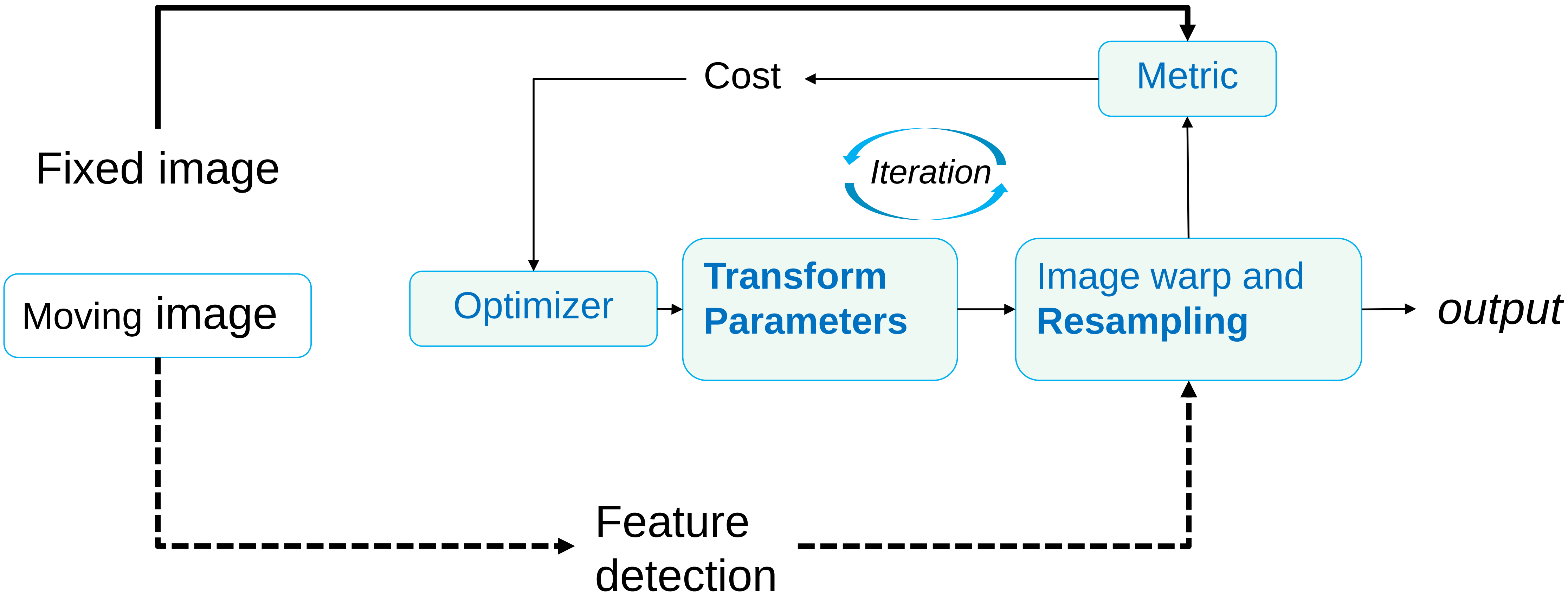
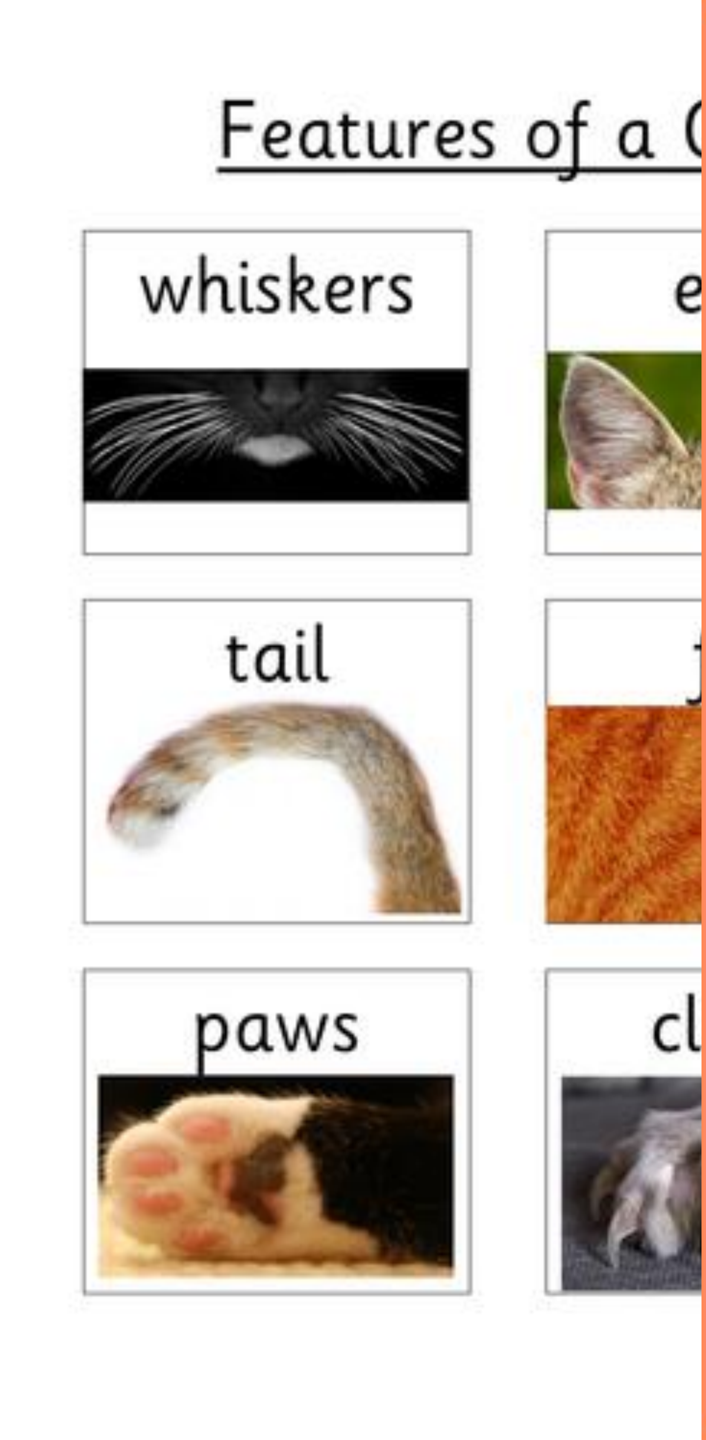
Previously

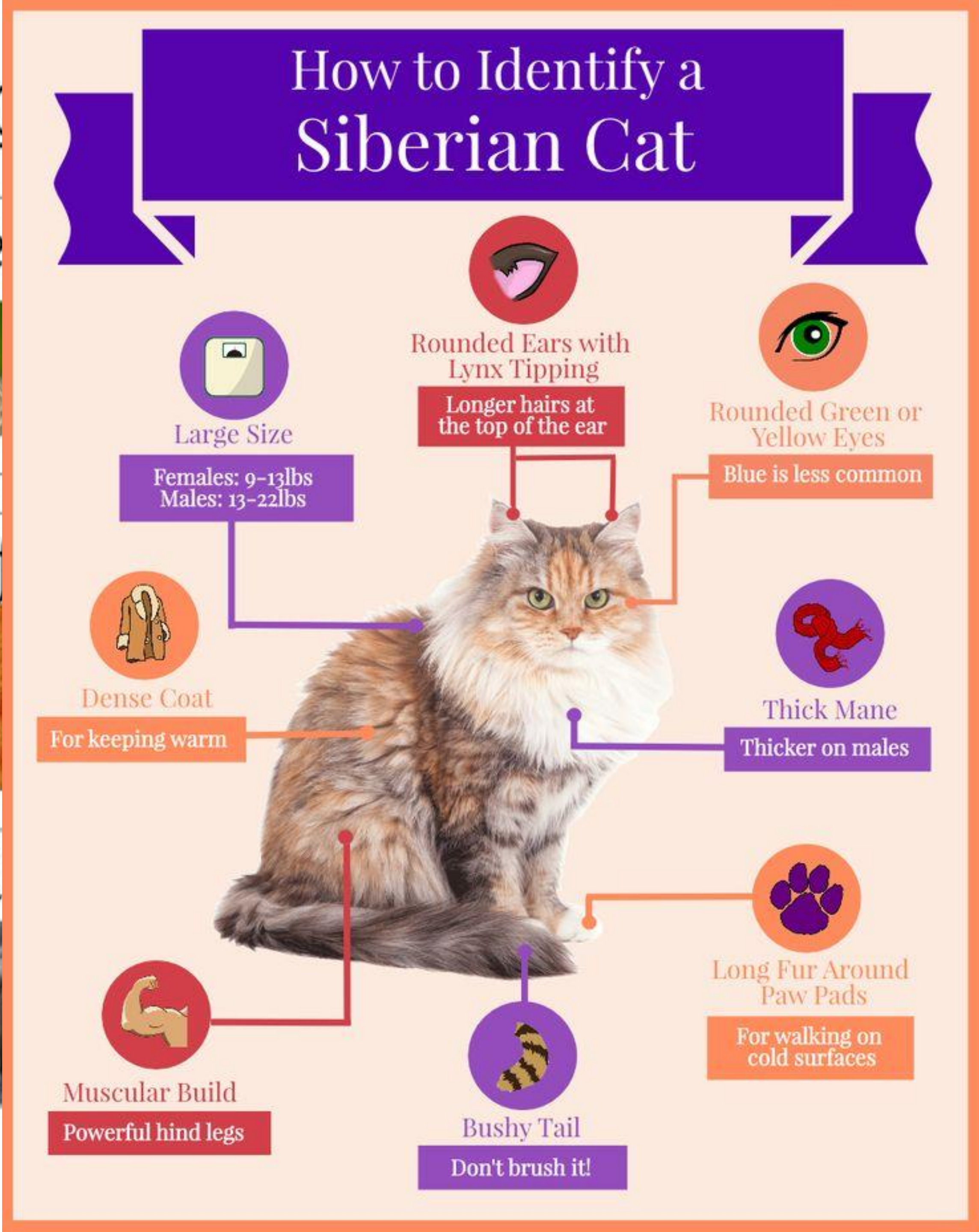

Cost
Metric
Iteration
Fixed image
Moving image
Optimizer
Transform Parameters
Image warp and Resampling
output
Feature detection

Outline

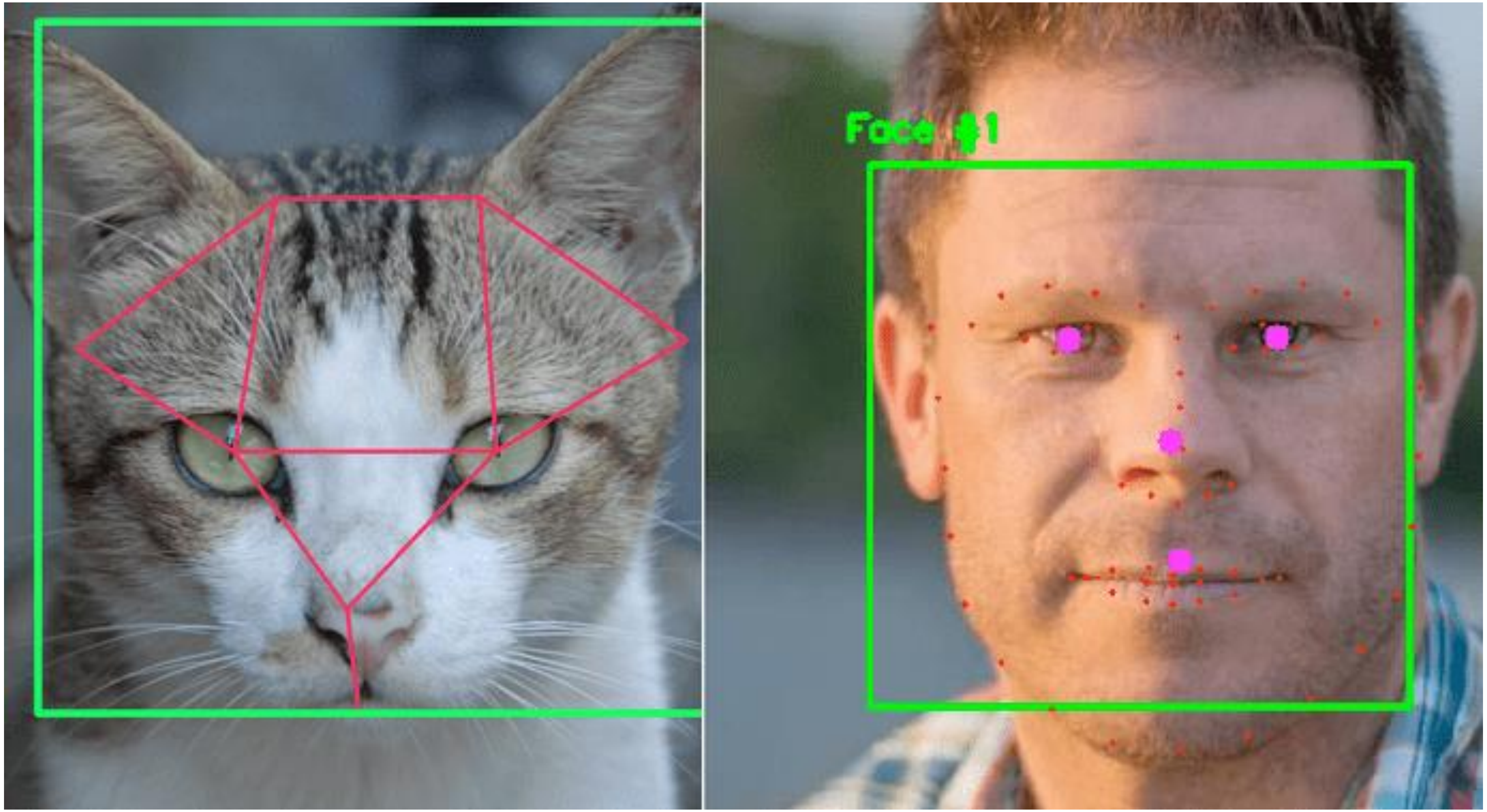
- Background
- Shape Feature
  - Boundary
  - Region
- Intensity Feature
  - Histogram-based
- Texture Feature
  - GLCM
- Radiomics

What is a feature?


Features of a
whiskers
tail
paws


How to Identify a Siberian Cat
Rounded Ears with Lynx Tipping
Longer hairs at the top of the ear
Large Size
Females: 9-13lbs
Males: 13-22lbs
Rounded Green or Yellow Eyes
Blue is less common
Dense Coat
For keeping warm
Thick Mane
Thicker on males
Long Fur Around Paw Pads
For walking on cold surfaces
Muscular Build
Powerful hind legs
Bushy Tail
Don't brush it!

Features



Background

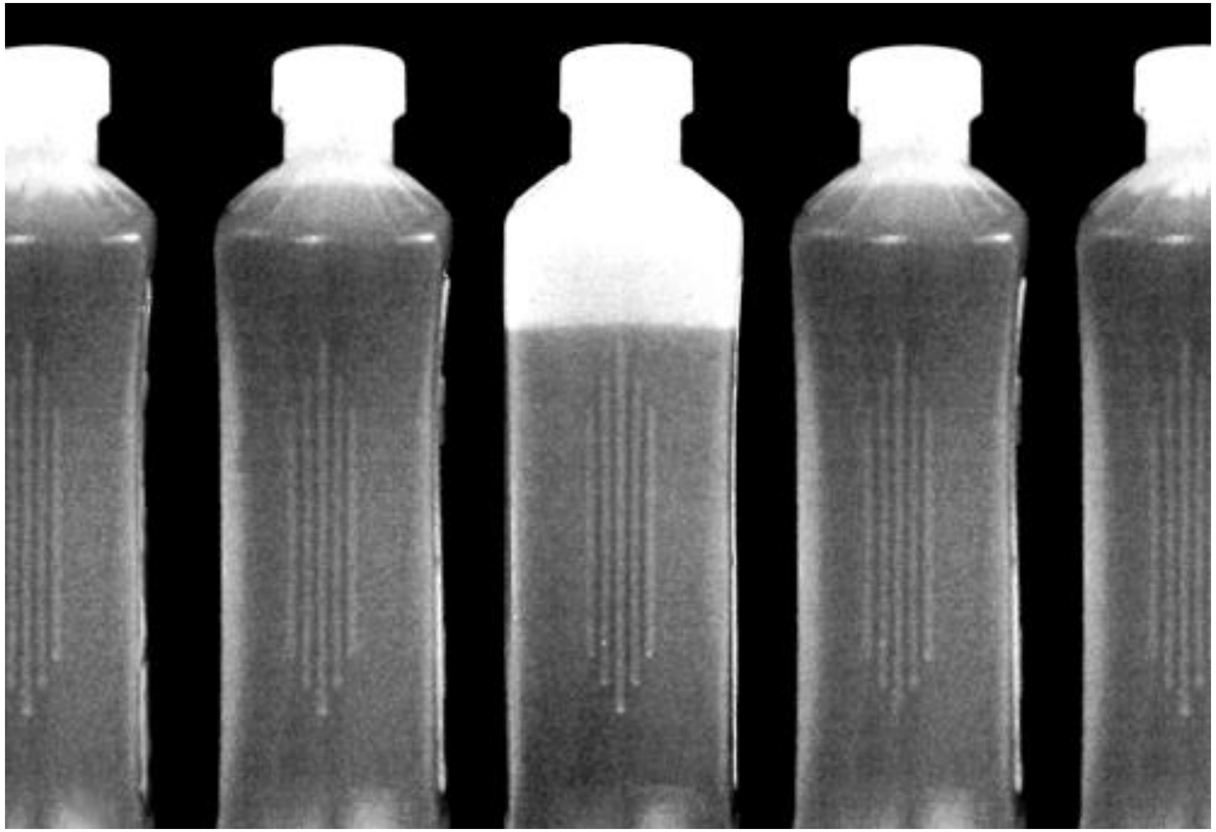
Segmentation → **Regions & Boundary** → **Feature extraction**

Feature extraction

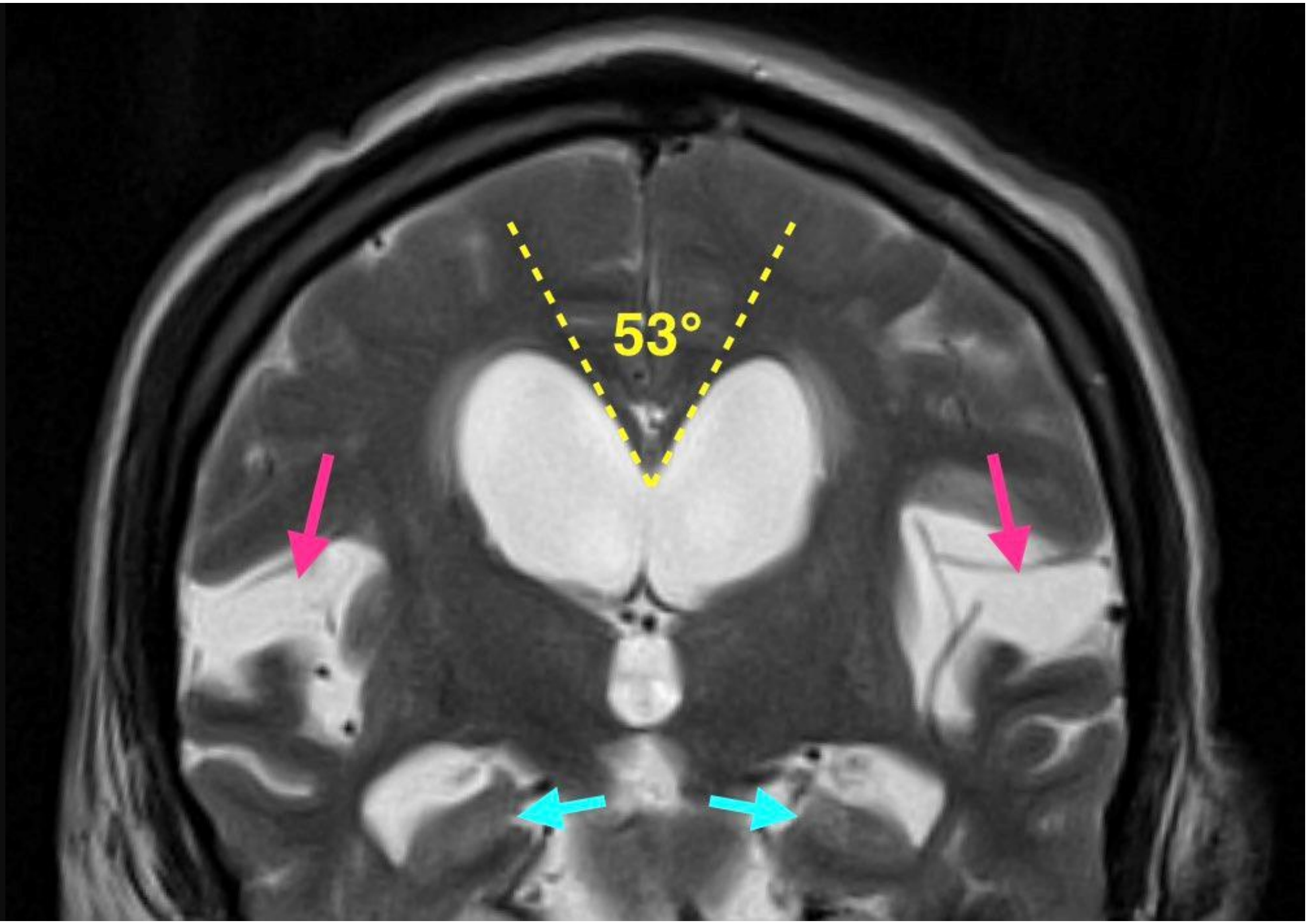
- **Feature detection**
  - Finding the features in a boundary, region, or image.
- **Feature description**
  - Assigns quantitative attributes to the detected features.

Features

- Feature example: area
- Local:
  - Each bottle
- Global:
  - All bottle
  - Total production



Features



Features

- There is no exact definition, but feature is a piece of information that we are interested about the content of image or region.

- Feature have these properties
  - Independent
  - Invariant
    - Affine transformations: {translation, reflection, rotation}
  - Covariant
    - Affine transformations: scaling

Features

- Shape Feature
  - Only consider the **spatial** information of ROI
  - Use **mask** to calculate
- Intensity Feature
  - Only consider the **intensity** information of ROI
  - Use **histogram** to calculate
- Texture Feature
  - Consider **both intensity and spatial** information of ROI
- Radiomics
  - Combination of many features

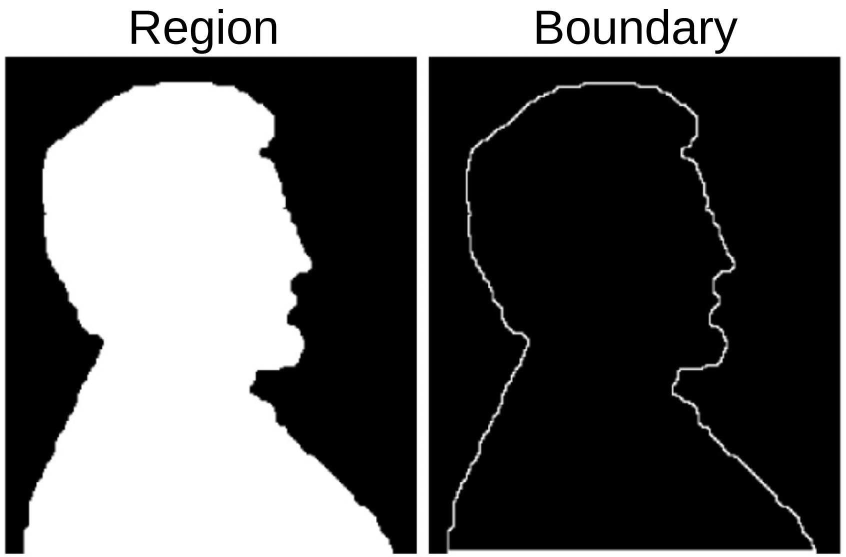
Shape Features

Shape Features

- Shape features are descriptors for the **size and shape** of the ROI. These features are **independent from the gray level intensity distribution** in the ROI.
- Shape features have 2D and 3D version, here we focus on 2D Shape Features.
- ROIs in fact are **region**, so the shape features can be divided into 2 groups.
  - **Boundary**: boundary, signature, skeleton

    derived form edge detection, watershed method…
  - **Region**

    derived form threshold, region-based methods…



Basic Boundary Descriptors

- The **length** of the boundary

  The ***number of pixels*** along a boundary is an approximation of its length.

- The **diameter** of a boundary B

$$diameter(B) = max_{i,j}[D(p_i, p_j)]$$

  where D is a distance measure and and p_i, p_j are points on the boundary.

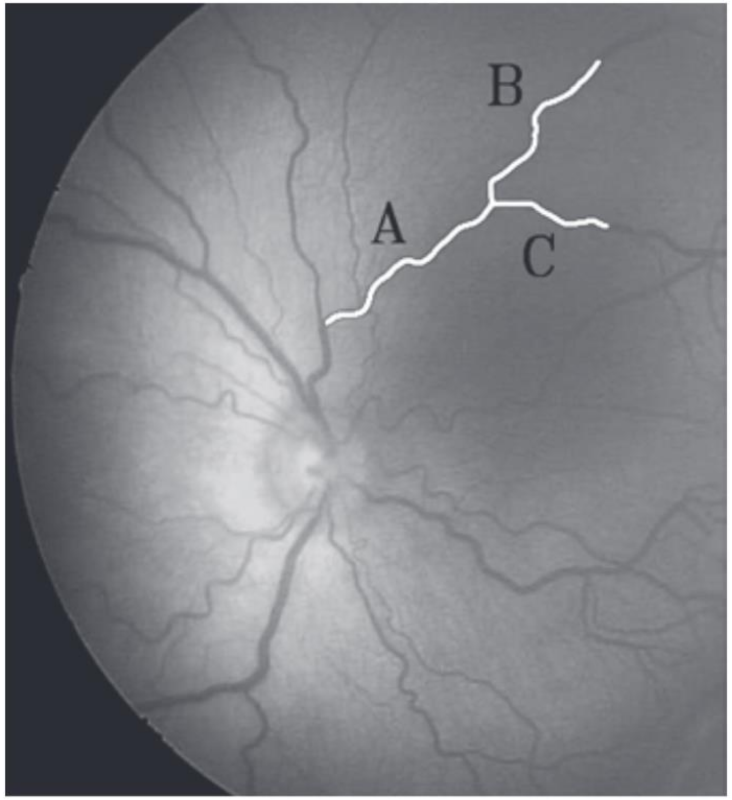
  - The **Major axis**: the value of the diameter and the orientation of a line segment connecting the two extreme points (x_1, x_2) that comprise the diameter.
  - The **Minor axis**: the line perpendicular to the major axis.

- The **eccentricity** of the boundary: the ratio of the focal length to the major axis
- The **curvature** of a boundary: the rate of change of slope

Basic Boundary Descriptors—Tortuosity

a b

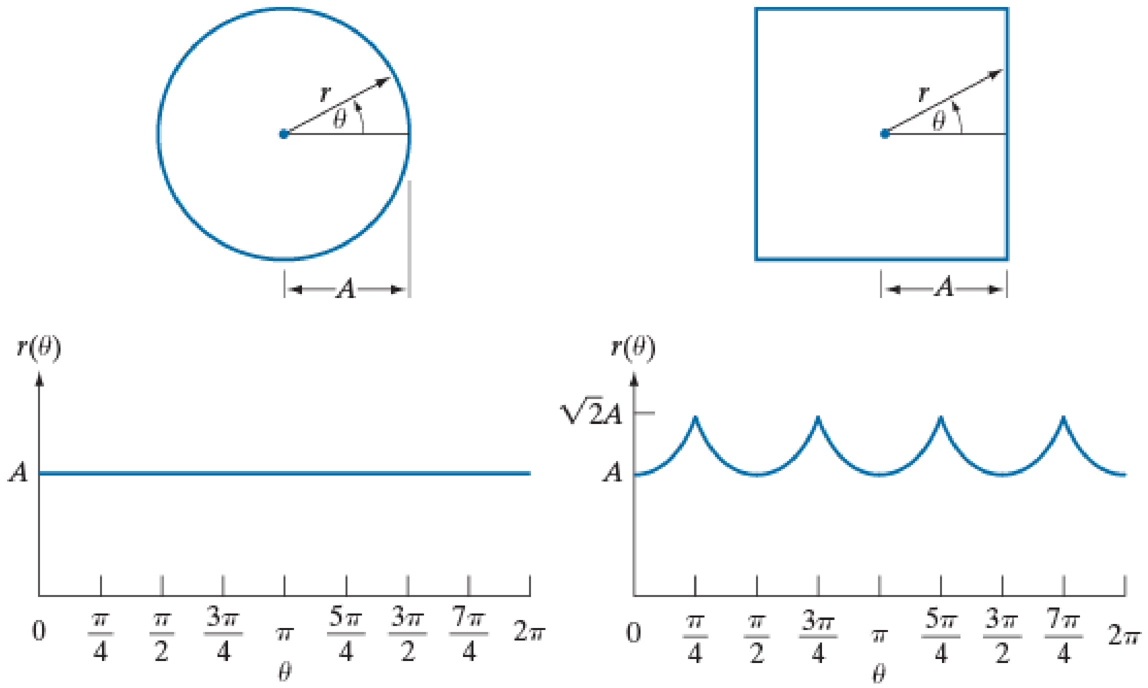
FIGURE 11.15
(a) Fundus image from a prematurely born baby with ROP. (b) Tortuosity of vessels A, B, and C. (Courtesy of Professor Ernesto Bribiesca, IIMAS-UNAM, Mexico.)

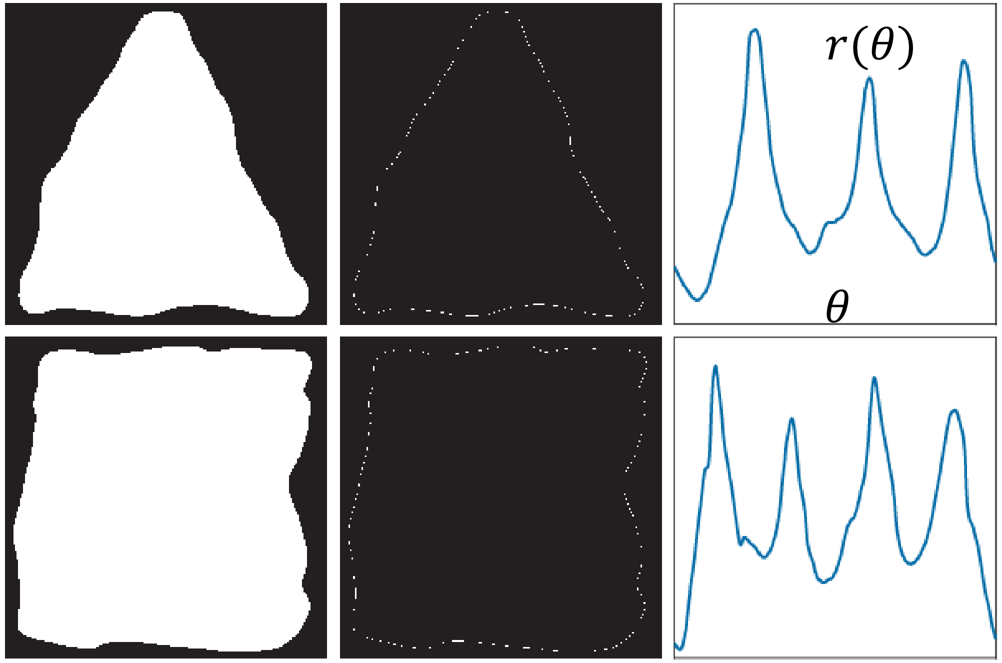


Curve	n	τ
A	50	2.3770
B	50	2.5132
C	50	1.6285

Basic Boundary Descriptors—Signatures

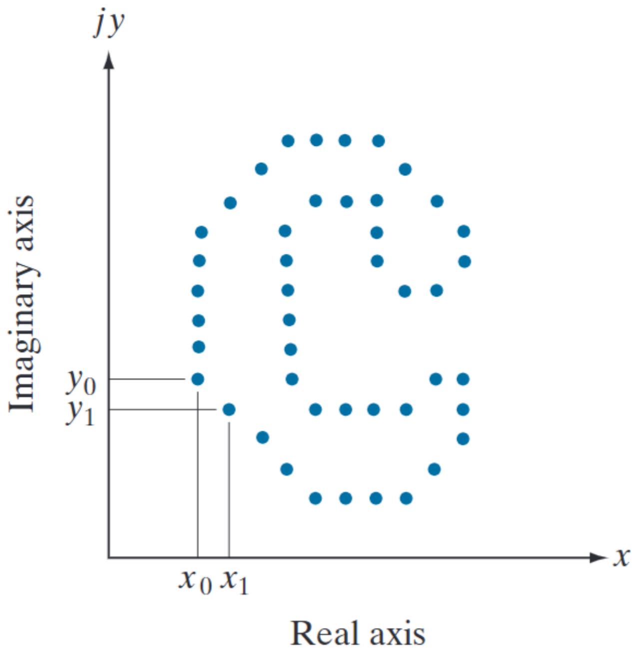
- A signature is a 1-D functional representation of a 2-D boundary and may be generated in various ways. One of the simplest is to plot the distance from the centroid to the boundary as a function of angle





The **number of prominent peaks** in the signatures is sufficient to differentiate between the shapes of the two objects

Fourier Descriptors



- A digital boundary in the xy-plane, consisting of ***K*** points.
- Starting at an arbitrary point (x_0, y_0), coordinate pairs $(\boldsymbol{x_0}, \boldsymbol{y_0}), (\boldsymbol{x_1}, \boldsymbol{y_1}), (\boldsymbol{x_2}, \boldsymbol{y_2}), \ldots\ (x_k, y_k)$ are encountered in traversing the boundary, say, in the counterclockwise direction.
- These coordinates can be expressed in the form：

$$x(k) = x_k \ and \ y(k) = y_k$$

- The boundary itself can be represented as:

$$s(k) = [x(k), y(k)] \quad k = 0,1,2, \ldots, K-1$$

- Moreover, each coordinate pair can be treated as a complex number: $s(k) = x(k) + jy(k)$

Fourier Descriptors:

The discrete Fourier transform (DFT) of $s(k)$:

$$a(u) = \sum_{k=0}^{K-1} s(k) e^{-j2\pi uk/K}$$

The inverse Fourier transform of these coefficients restores $s(k)$: $s(k) = \frac{1}{K}\sum_{u=0}^{K-1} a(u) e^{j2\pi uk/K}$

Fourier Descriptors

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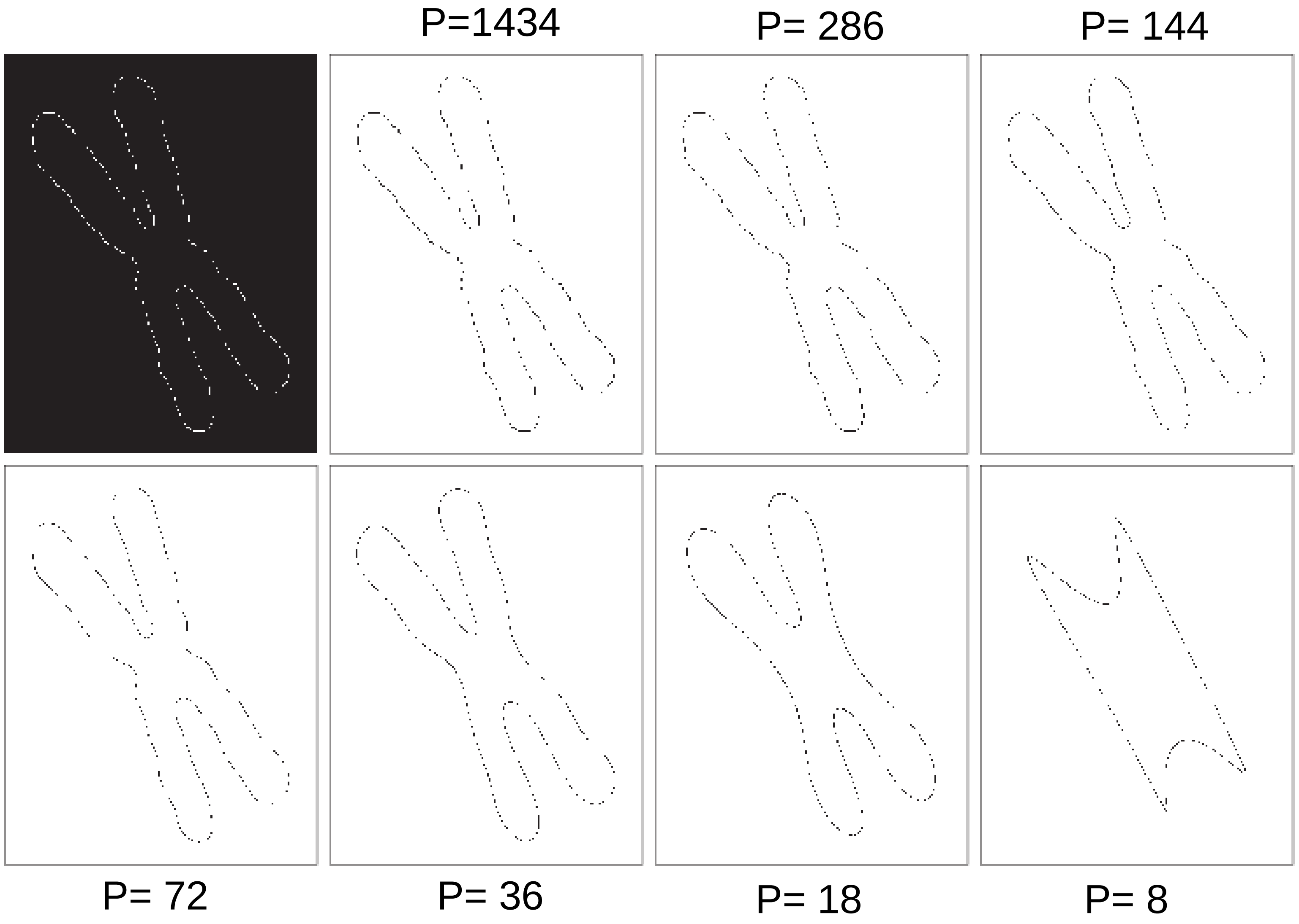
- We can use the **first P coefficients** to calculate approximated $\hat{s}(k)$

$$\hat{s}(k) = \frac{1}{K}\sum_{u=0}^{P-1} a(u) e^{j2\pi uk/K}. \quad \text{for k = 0,1,2,…,K-1}$$

  Although only P terms are used to obtain each component of $\hat{s}(k)$, parameter k still ranges from 0 to $K-1$. That is, the same number of points exists in the approximate boundary, but not as many terms are used in the reconstruction of each point.

- High-frequency components account for fine detail, and low-frequency components determine overall shape.

Fourier Descriptors



Boundary of a human chromosome (2868 points). (b)–(h) Boundaries reconstructed using 1434, 286, 144, 72, 36, 18, and 8 Fourier descriptors, respectively.

Basic Region Descriptors

- **Perimeter** p, the pixel number of the region's boundary.
- **Area** A, the pixel number of the region.
- **Compactness**, defined as the perimeter p squared over the area A

$$\text{compactness} = \frac{p^2}{A}$$

independent of orientation and translation

- **Circularity**, closely related to compactness

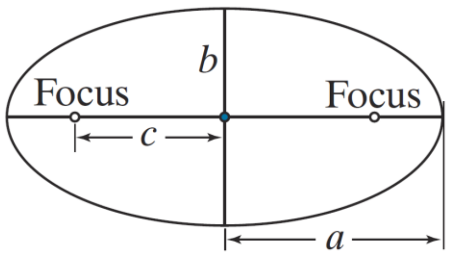
$$\text{circularity} = \frac{4\pi A}{p^2}$$

Basic Boundary Descriptors—Eccentricity

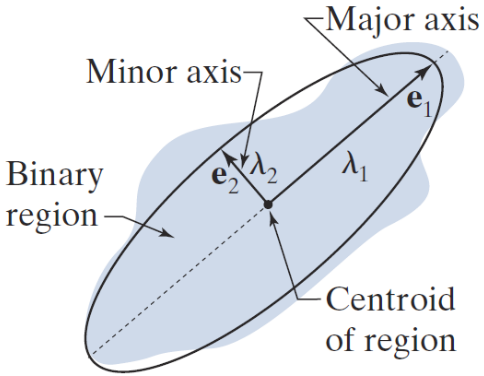
- **Eccentricity**, based on an ellipse

$$\text{eccentricity} = \frac{c}{a} = \frac{\sqrt{a^2 - b^2}}{a} = \sqrt{1 - (b/a)^2}\,, (a \geq b)$$

$$c^2 = a^2 - b^2$$



- **Eccentricity** approximates 2-D data by an elliptical region



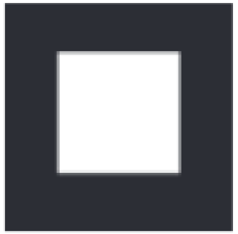
$\mathbf{e}_1$ λ_1 and $\mathbf{e}_2$ λ_2 are the eigenvectors and corresponding eigenvalues of the covariance matrix of the coordinates of the region

The principal axes are the eigenvectors of the covariance matrix, C

$$\boldsymbol{C} = \frac{1}{K-1}\sum_{k=1}^{K}(\boldsymbol{z_k} - \bar{\boldsymbol{z}})(\boldsymbol{z_k} - \bar{\boldsymbol{z}})^T \qquad \text{where } \bar{\boldsymbol{z}} = \frac{1}{K}\sum_{k=1}^{K}\boldsymbol{z_k}$$

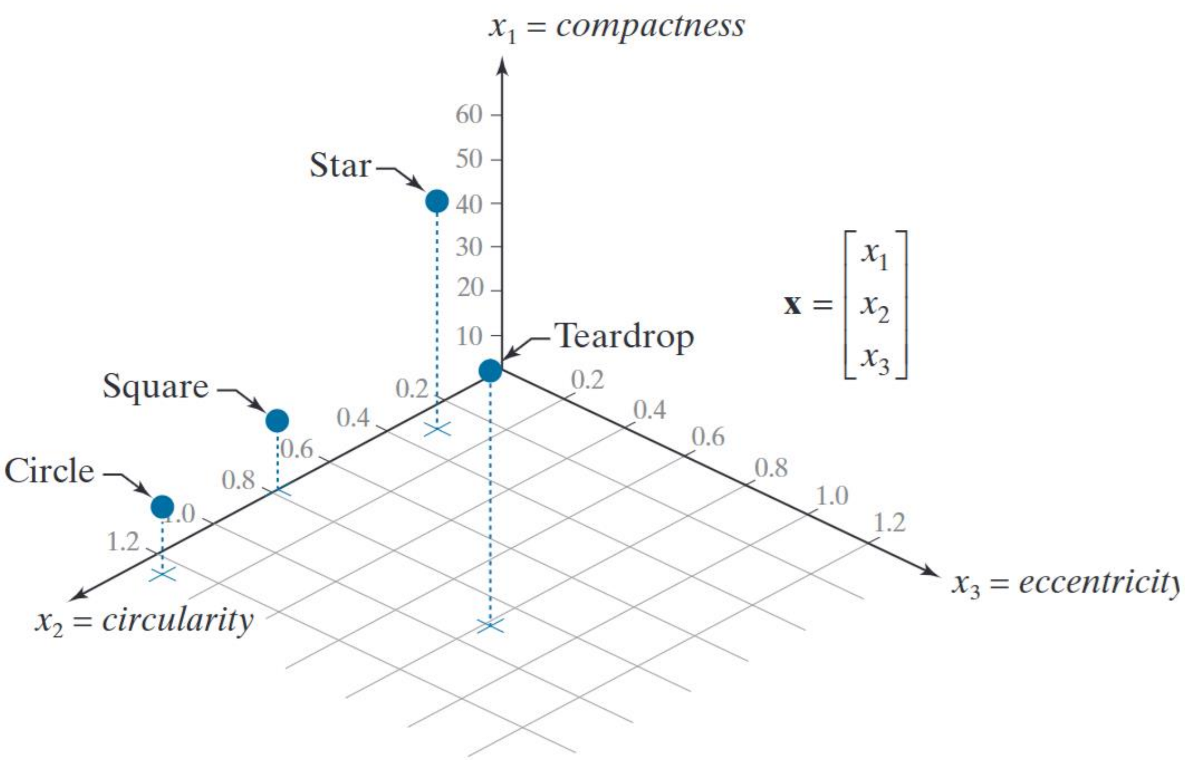
K: Total number of points

Basic Region Descriptors

Descriptor				
Compactness	10.1701	42.2442	15.9836	13.2308
Circularity	1.2356	0.2975	0.7862	0.9478
Eccentricity	0.0411	0.0636	0	0.8117

- The **circle** is highly compact and circular with low eccentricity
- The **star** is less compact and less circular than the other shapes
- The **teardrop** region has by far the largest eccentricity, but it is harder to differentiate from the other shapes using compactness or circularity

Feature Vectors

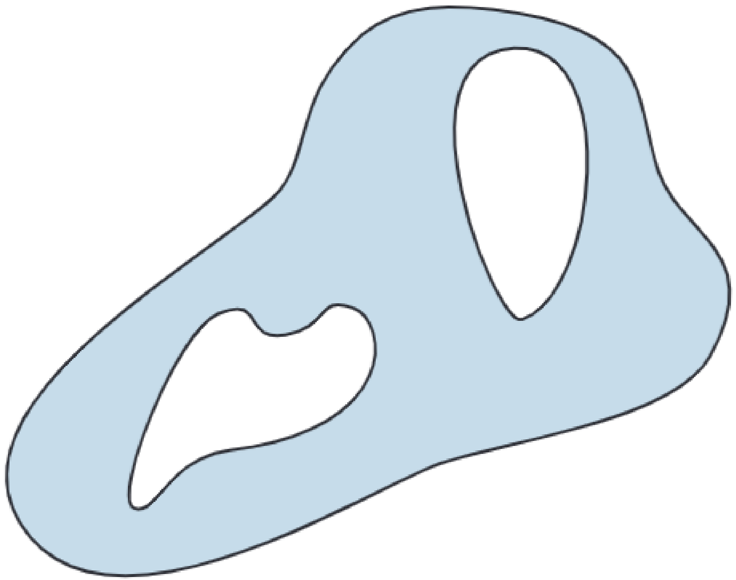


When having **multiple** feature, Feature vector can assess the similarity between different image.

- The circle and square are much more similar
- The star and teardrop objects are far from each other and from the circle and square
- Feature space will play an important role in image pattern classification

Topological Descriptors

Topological properties do not depend on the notion of distance or any properties implicitly based on the concept of a distance measure



- The number of holes in the region
  - Will **not** be affected by a stretching or rotation transformation

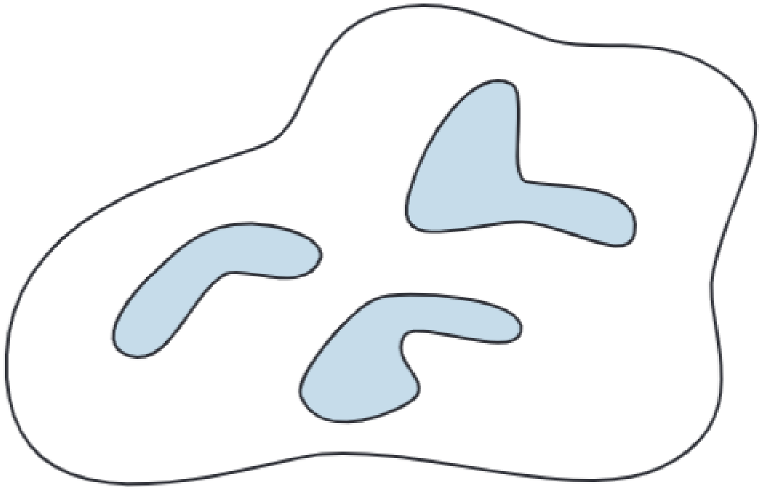
Topological Descriptors

- The number of connected components of an image or region

Euler number $E = C - H$

connected components

hole



Example of Euler number

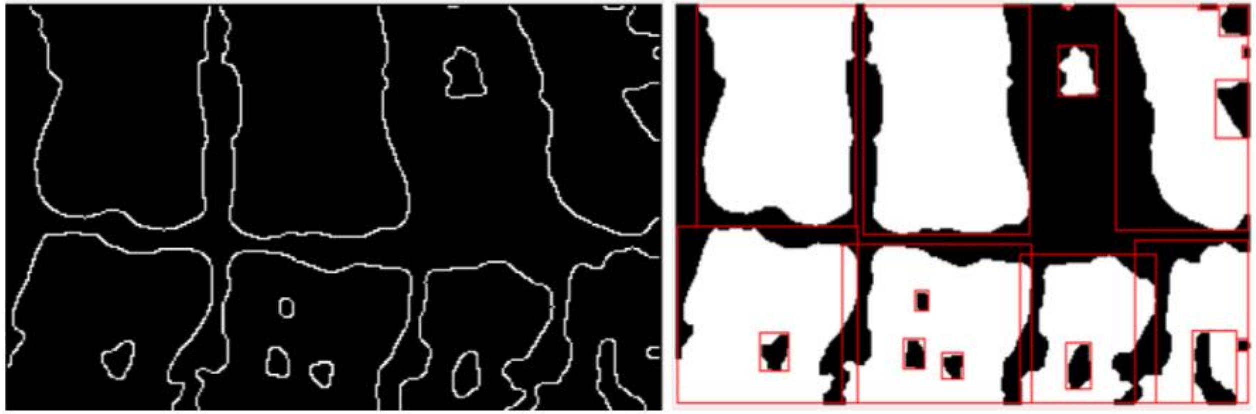


Figure (13) Teeth edge detection

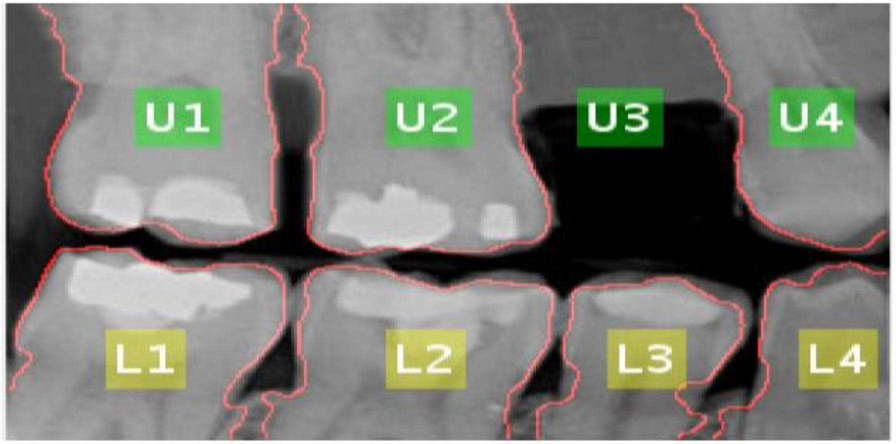


Figure (14) Edge coloring, objects removing and labeling with missing tooth

Edge Detection and Features Extraction for Dental X-Ray. DOI:10.30684/etj.34.13A.5

Shape features

- Boundary
  - Basic: length, diameter
  - Signature
  - Fourier
- Region
  - Basic: Compactness, Circularity…
  - Topological: Euler number
- Feature vector

Intensity Features

Intensity Features

- Intensity feature describe the distribution of **pixel/voxel intensities** within the Region of Interest (ROI) through commonly used and basic metrics.
- Only care pixel/voxel's intensities distribution within ROI, **no spatial information**.
- For digital image $f(x, y)$, the ROI is N.

Energy

$$Energy = \sum_{(x,y)\in N} f(x, y)^2$$

Mean

$$\bar{f} = \frac{1}{|N|} \sum_{(x,y)\in N} f(x, y)$$

Median

$$Median = median\ intensity\ in\ N$$

Intensity Features

- For digital image $f(x,y)$, the ROI is N.

Maximum/Minimum

$$\mathrm{Max} = \max_{(x,y)\in N} f(x,y) \qquad \mathrm{Min} = \min_{(x,y)\in N} f(x,y)$$

Range

$$range = Max(f) - Min(f)$$

Standard deviation

$$\sigma = \sqrt{\frac{1}{|N|}\sum_{(x,y)\in N}\left(f(x,y) - \bar{f}\right)^2}$$

Variance

$$Variance = \frac{1}{|N|}\sum_{(x,y)\in N}\left(f(x,y) - \bar{f}\right)^2 = \sigma^2$$

Example: Intensity feature

Region no. (from top)	Ratio of lights per region to total lights
1	0.204
2	0.640
3	0.049
4	0.107

Intensity Features

Histogram-based feature

- z is the variable denoting pixel intensity
- $p(z_i), i = 1,2,\ldots,L-1$ be the corresponding normalized histogram.

- **Uniformity**

$$U(z) = \sum_{i=0}^{L-1} p^2(z_i)$$

U reach maximum for a constant image

Low uniformity

High uniformity

- **Average entropy**

$$e(z) = -\sum_{i=0}^{L-1} p\,(z_i) log_2 p(z_i)$$

Entropy reaches minimum for a constant image

Intensity Features

n^{th} **moment** of mean value

$$\mu_n = \frac{1}{|N|}\sum_{(x,y)\in N}\left(f(x,y) - \bar{f}\right)^n$$

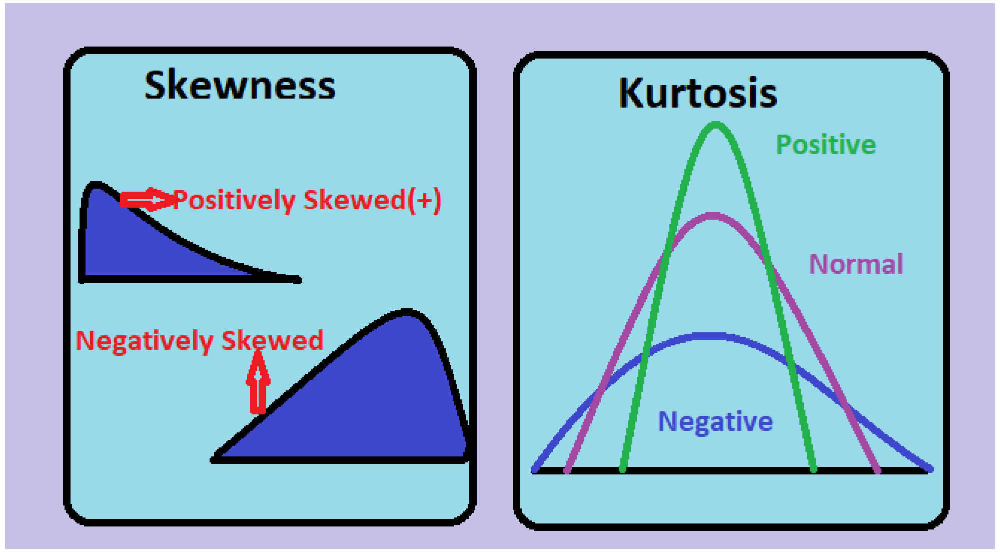
μ_2: **contrast** (variance); μ_3: **asymmetry** (skewness); μ_4: **peakedness** (kurtosis)

- **Skewness** (asymmetry of data around its mean)

$$Skewness = \frac{\mu_3}{\sigma^3} = \frac{\frac{1}{|N|}\sum_{(x,y)\in N}(f(x,y)-\bar{f})^3}{\sigma^3}$$

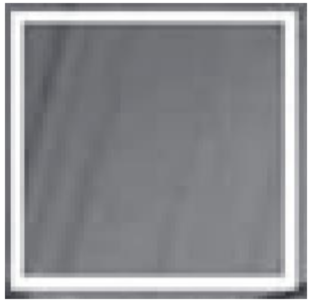
- **Kurtosis** (how much data is in the "tails")

$$Kurtosis = \frac{\mu_4}{\sigma^4} = \frac{\frac{1}{|N|}\sum_{(x,y)\in N}(f(x,y)-\bar{f})^4}{\sigma^4}$$



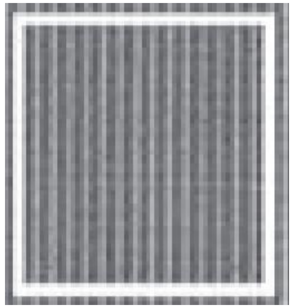
Intensity Features

Smooth



Coarse

Regular



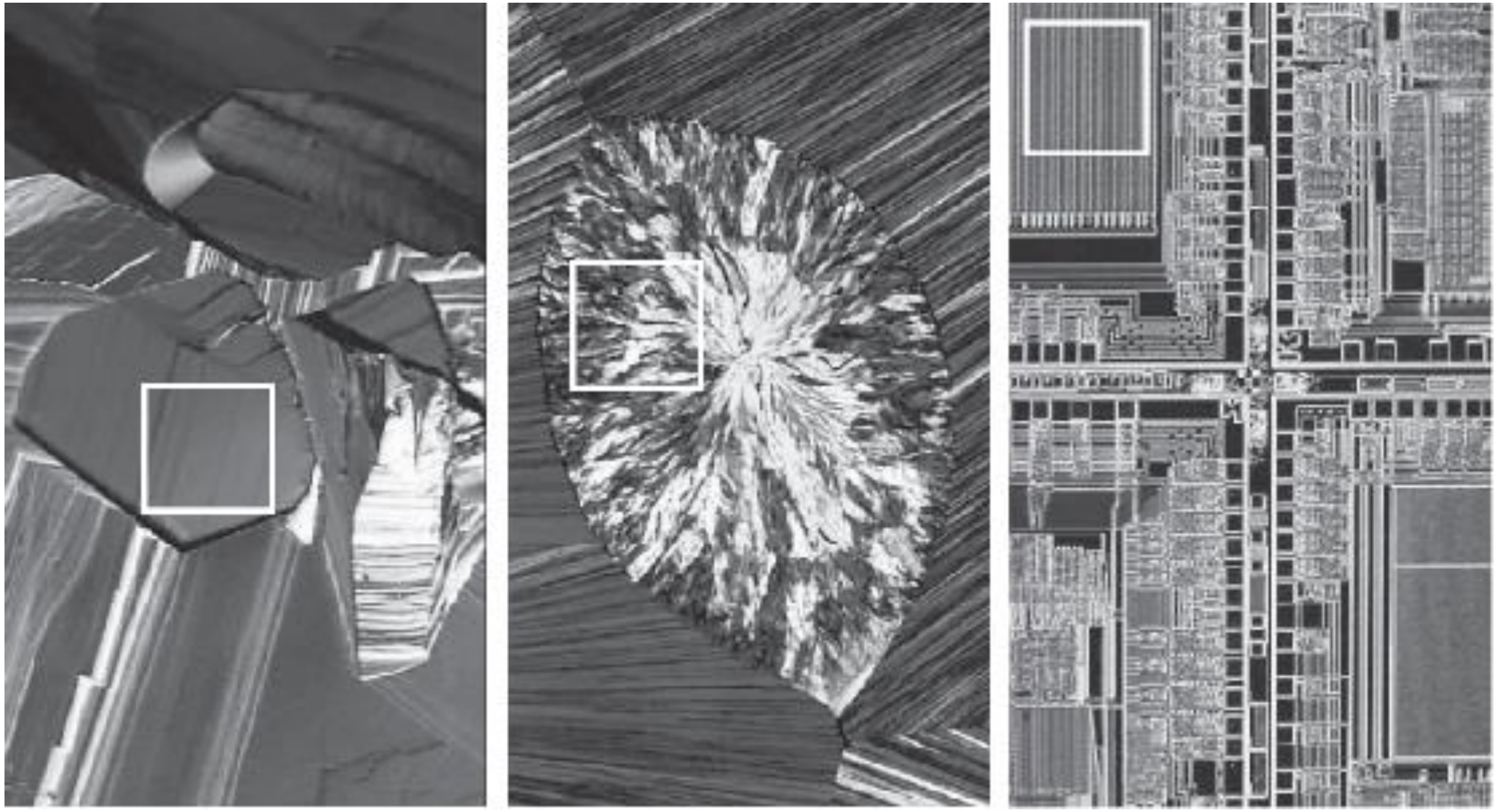
$$R_z = 1 - \frac{1}{1 + \sigma^2(z)}$$

Texture	Mean	Standard deviation	*R* (normalized)	3rd moment	Uniformity	Entropy
Smooth	82.64	11.79	0.002	−0.105	0.026	5.434
Coarse	143.56	74.63	0.079	−0.151	0.005	7.783
Regular	99.72	33.73	0.017	0.750	0.013	6.674

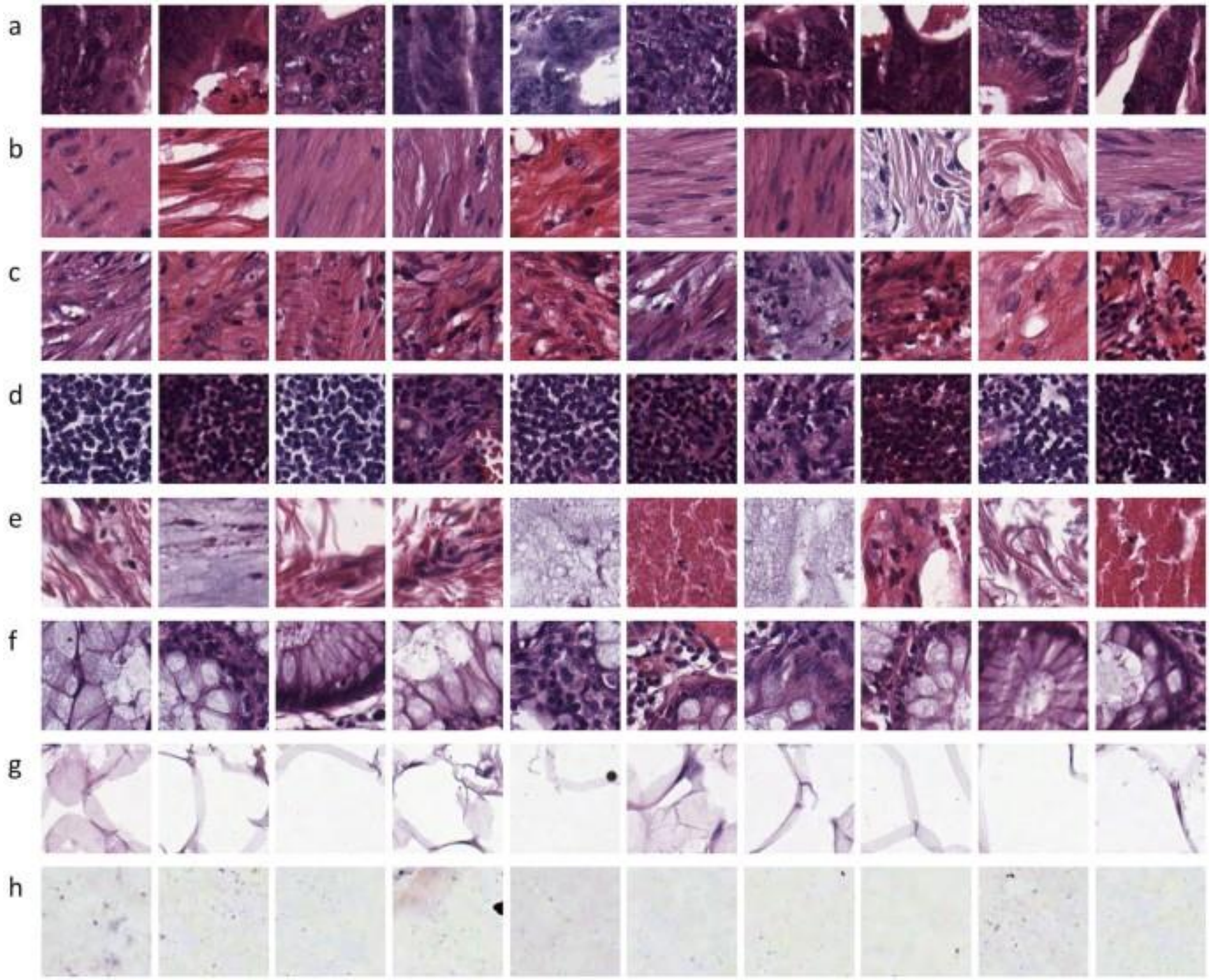
Texture Features

Texture

- No formal definition of texture exists
  - intuitively, smoothness, coarseness, and regularity



Texture



Spatial Information: Co-occurrence matrix

1. Set the image intensity level K.
2. Set the direction for co-occurrence.
3. The matrix G is a matrix with size $K \times K$.
4. Pixel and it's co-occurrence pixel (depended by the direction) intensity decides G's index.

1	1	7	5	3	2
5	1	6	1	2	5
8	8	6	8	1	2
4	3	4	5	5	1
8	7	8	7	6	2
7	8	6	2	6	2

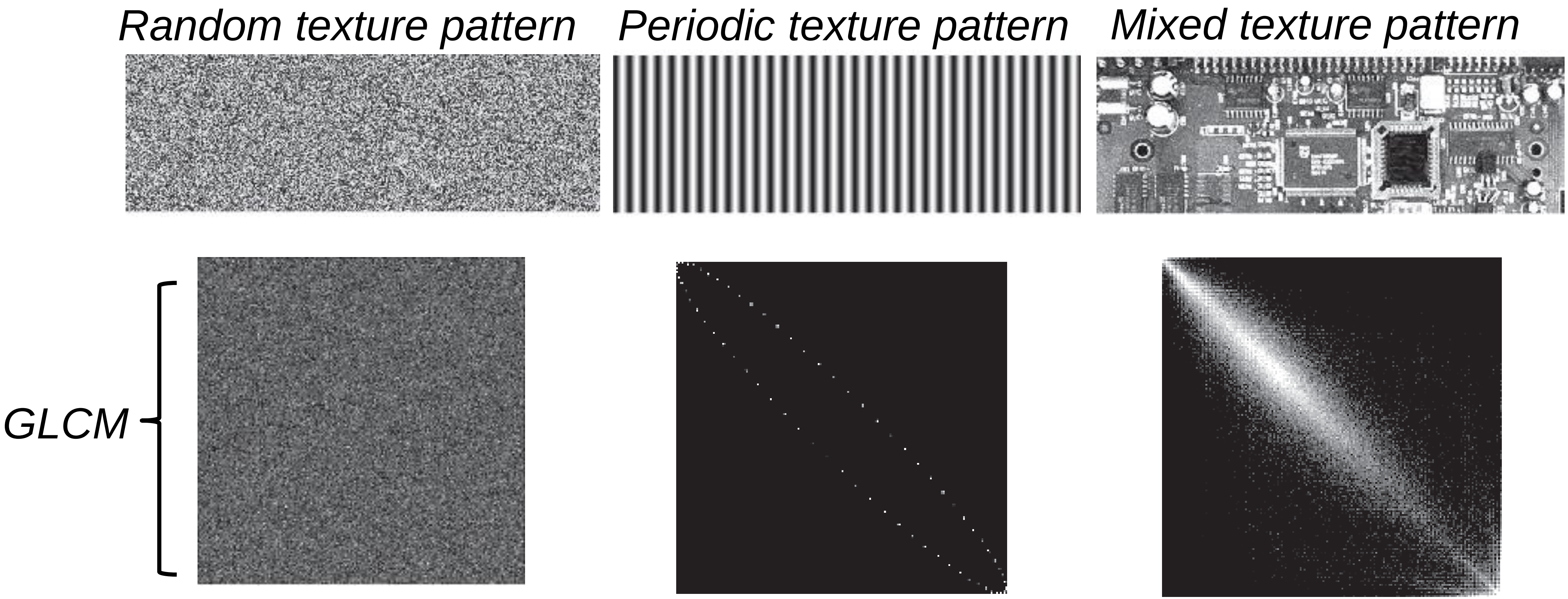
Image f

	1	2	3	4	5	6	7	8
1	1	2	0	0	0	1	1	0
2	0	0	0	0	1	1	0	0
3	0	1	0	1	0	0	0	0
4	0	0	1	0	1	0	0	0
5	2	0	1	0	1	0	0	0
6	1	3	0	0	0	0	0	1
7	0	0	0	0	1	1	0	2
8	1	0	0	0	0	2	2	1

Co-occurrence matrix **G**

pixel pairs with intensities

Spatial Information: Co-occurrence matrix



- For reducing computations the intensities can be quantified into a few bands in order to keep the size of G manageable.

Spatial Information: Co-occurrence matrix

- Position operator Q definition:

  E.g.- one pixel immediately to the right?

  - one pixel to the right and one pixel above ?

- The probability that a pair of points satisfying Q will have values (z_i, z_j) in G:

$$p_{ij} = \frac{g_{ij}}{n}$$

- These probabilities are in the range $[0, 1]$ and their sum is 1:

$$\sum_{i=1}^{K} \sum_{j=1}^{K} p_{ij} = 1$$

Spatial Information: Co-occurrence matrix

Descriptors used for characterizing co-occurrence matrices of size $K \times K$

E.g. Maximum probability, correlation, contrast, uniformity(Energy), Homogeneity

- Correlation $$\sum_{i=0}^{K}\sum_{j=0}^{K}\frac{(i-m_r)(j-m_c)p_{ij}}{\sigma_r\sigma_c}$$

where $m_r = \sum_{i=1}^{K} i \sum_{j=1}^{K} p_{ij}$ $\quad m_c = \sum_{j=1}^{K} j \sum_{i=1}^{K} p_{ij}$

$$\sigma_r^2 = \sum_{i=1}^{K}(i-m_r)^2\sum_{j=1}^{K}p_{ij} \qquad \sigma_c^2 = \sum_{j=1}^{K}(j-m_c)^2\sum_{i=1}^{K}p_{ij}$$

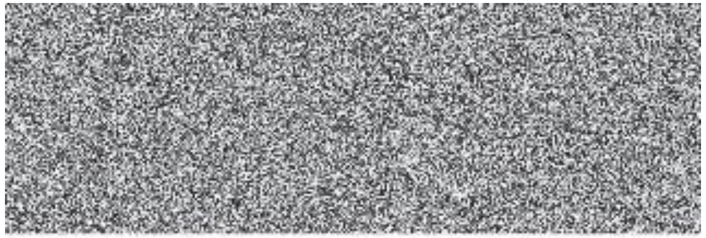
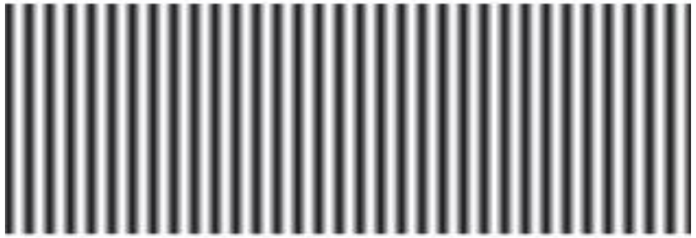
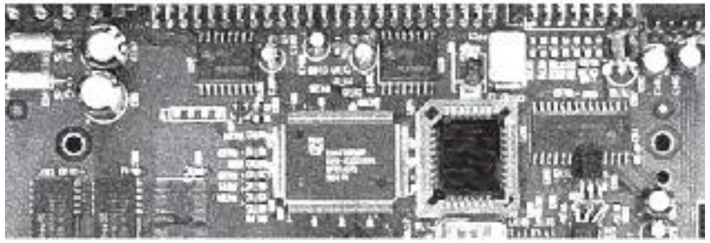
Along rows

Along columns

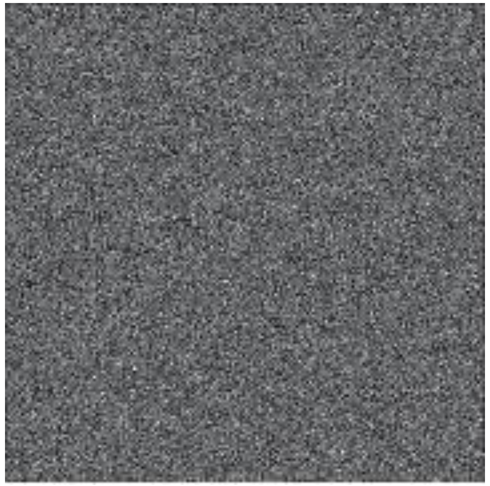
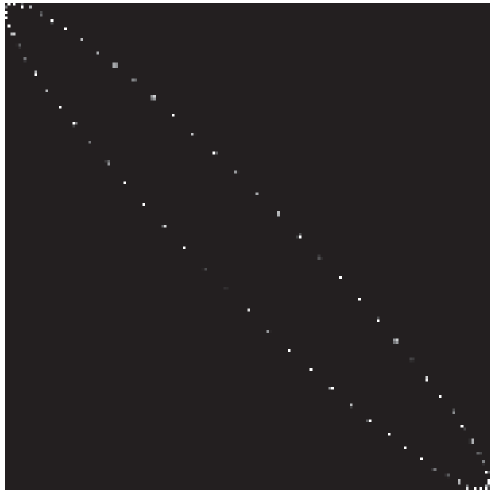
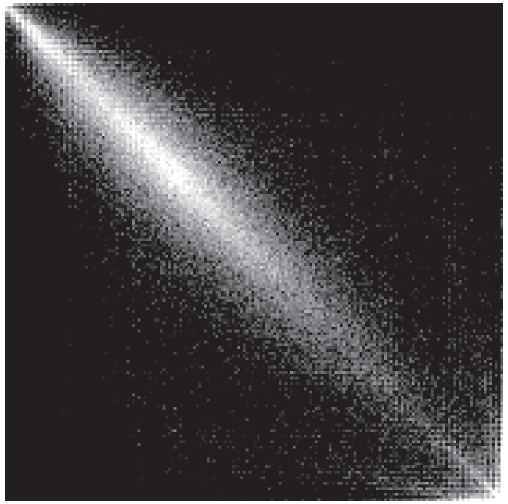
Descriptors used for characterizing co-occurrence matrices

Descriptor	Explanation	Formula
Maximum probability	Measures the strongest response of G. The range of values is $[0,1]$.	$\max_{i,j}(p_{ij})$
Correlation	A measure of how correlated a pixel is to its neighbor over the entire image. The range of values is 1 to -1 corresponding to perfect positive and perfect negative correlations. This measure is not defined if either standard deviation is zero.	$\sum_{i=1}^{K}\sum_{j=1}^{K}\frac{(i-m_r)(j-m_c)p_{ij}}{\sigma_r\sigma_c}$ $\sigma_r \neq 0, \sigma_c \neq 0$
Contrast	A measure of intensity contrast between a pixel and its neighbor over the entire image. The range of values is 0 (when G is constant) to $(K-1)^2$	$\sum_{i=1}^{K}\sum_{j=1}^{K}(i-j)^2 p_{ij}$
Uniformity (also called Energy)	A measure of uniformity in the range $[0,1]$. Uniformity is 1 for a constant image.	$\sum_{i=1}^{K}\sum_{j=1}^{K}p_{ij}^2$
Homogeneity	Measures the spatial closeness to the diagonal of the distribution of elements in G. The range of values is $[0,1]$, with the maximum being achieved when G is a diagonal matrix.	$\sum_{i=1}^{K}\sum_{j=1}^{K}\frac{p_{ij}}{1+\lvert i-j\rvert}$
Entropy	Measures the randomness of the elements of G. The entropy is 0 when all p_{ij}'s are 0, and is maximum when the p_{ij}'s are uniformly distributed. The maximum value is thus $2\log_2 K$.	$-\sum_{i=1}^{K}\sum_{j=1}^{K}p_{ij}\log_2 p_{ij}$

Descriptors used for characterizing co-occurrence matrices

GLCM

Normalized Co-occurrence Matrix	Maximum Probability	Correlation	Contrast	Uniformity	Homogeneity	Entropy
$\mathbf{G}_1/n_1$	0.00006	–0.0005	10838	0.00002	0.0366	15.75
$\mathbf{G}_2/n_2$	0.01500	0.9650	00570	0.01230	0.0824	06.43
$\mathbf{G}_3/n_3$	0.06860	0.8798	01356	0.00480	0.2048	13.58

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COLLEGE OF BIOMEDICAL ENGINEERING & INSTRUMENT SCIENCE
ZHEJIANG UNIVERSITY

Texture

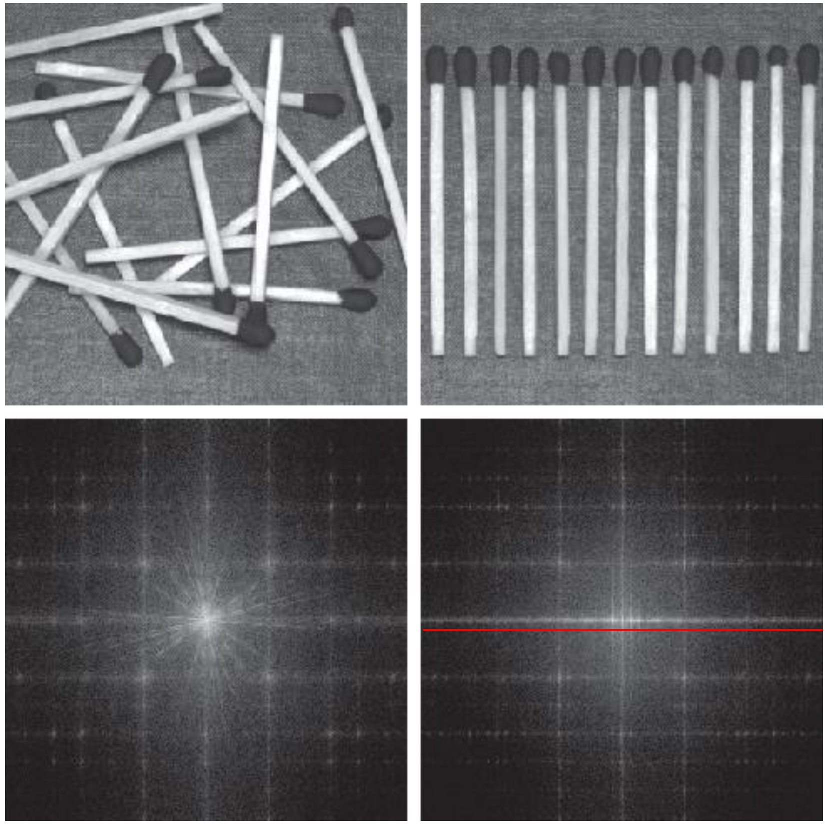
- **Statistical Approaches(first order, GLCM)**
- **Moment Invariants Approaches(Hu moments)**
- **Spectral Approaches(Fourier)**

Why spectral?

(1) Peaks in the spectrum give the principal direction of the texture patterns;

(2) The location of the peaks in the frequency plane gives the fundamental spatial period of the patterns;

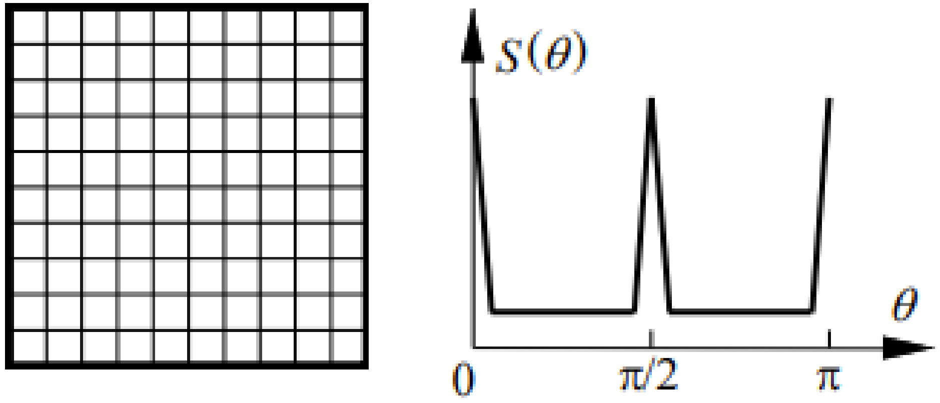
(3) Eliminating any periodic components via filtering leaves nonperiodic image elements, which can then be described by statistical techniques.

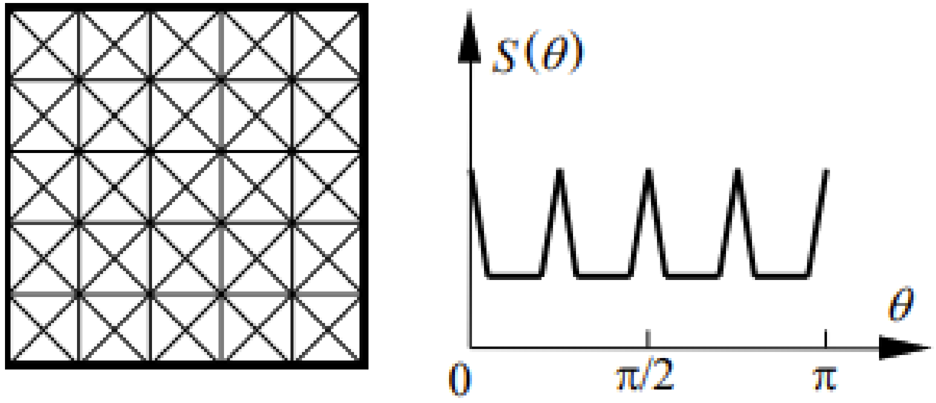


Spectral Texture

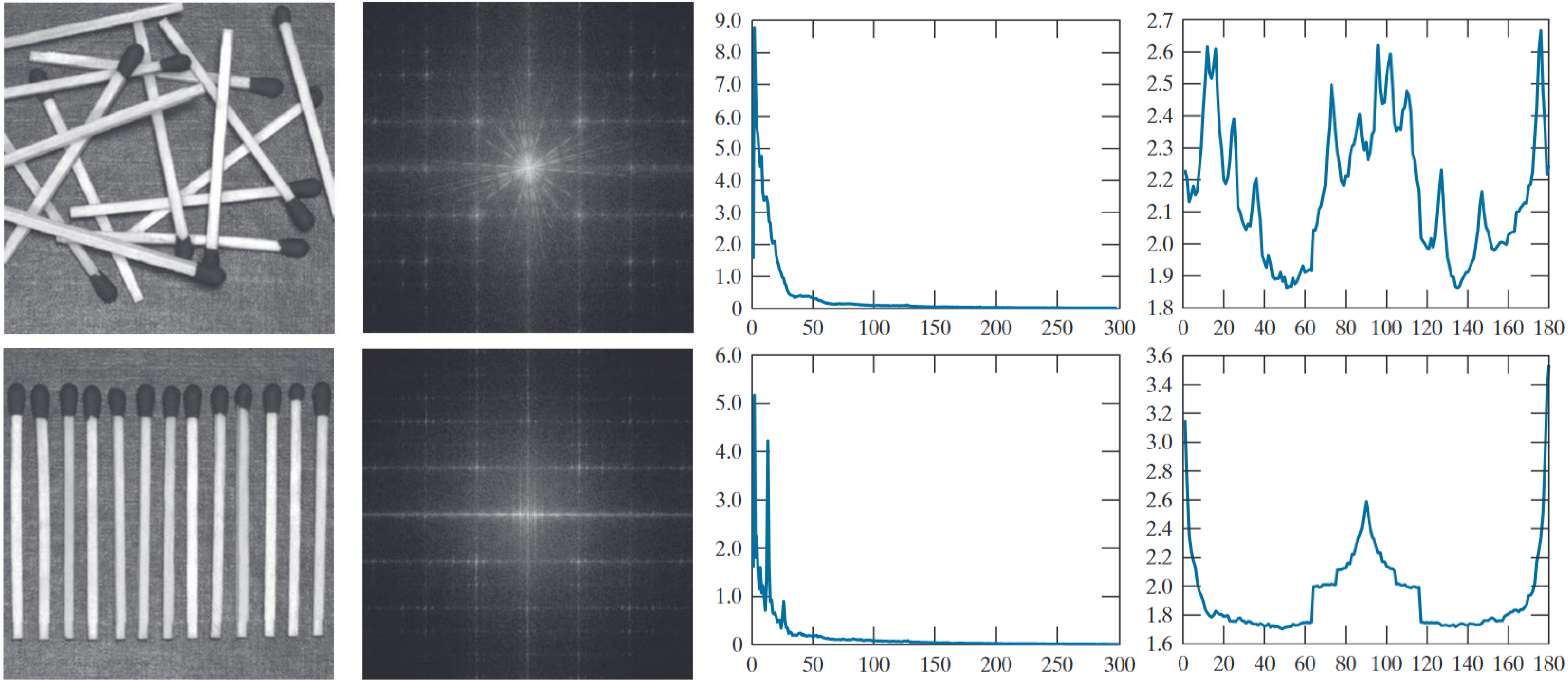
Spectral function in polar coordinate: $[S(u), S(v)] \rightarrow [S(r), S(\theta)]$

$$S(r) = \sum_{\theta=0}^{\pi} S_\theta(r) \qquad S(\theta) = \sum_{r=1}^{R} S_r(\theta)$$





Spectral Texture



$S(r)$ $S(\theta)$

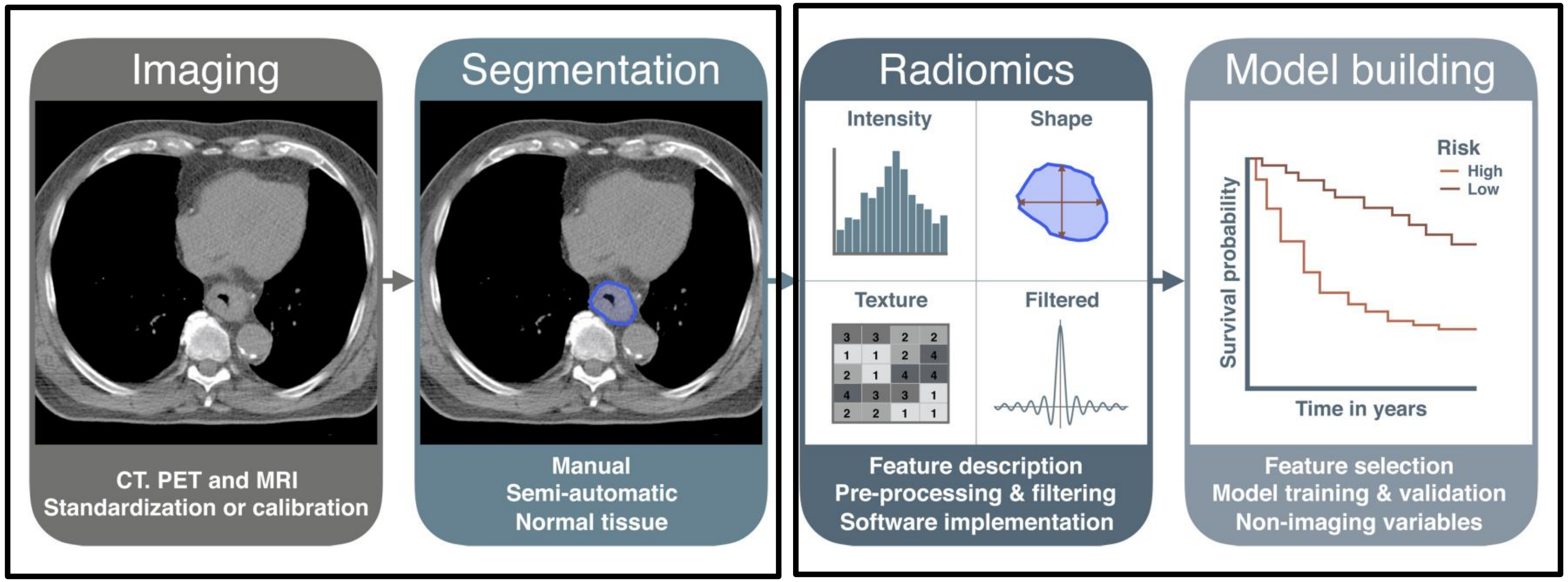
Texture features

- First order Approaches
  - Statistical
  - Histogram based
- GLCM: Correlation, Contrast…
- Spectral Approaches: Fourier

Radiomics

Introduction to Radiomics

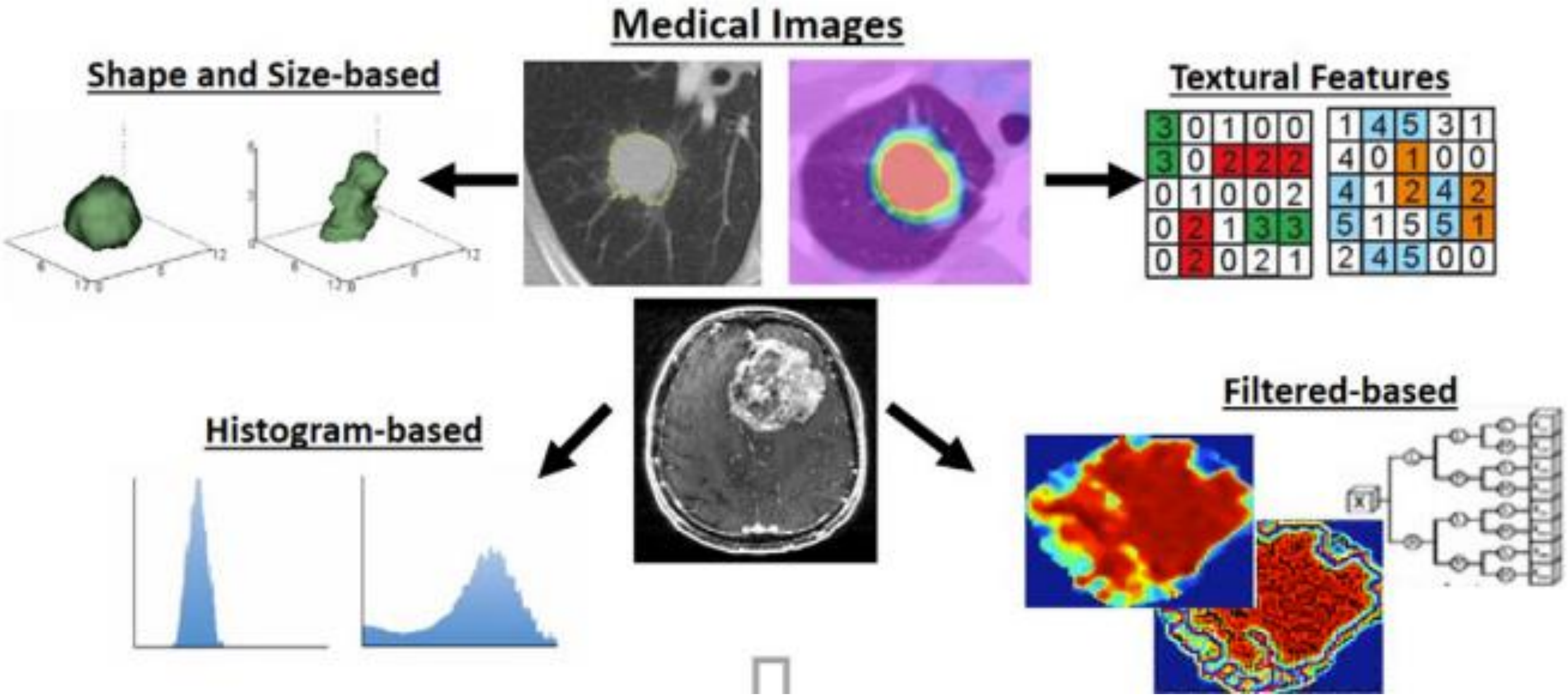
- Radiomics is a method to extract quantitative features(**-mics**) from an region of interest on medical images(**radio-**) for further analysis.
- Radiomics converses digital medical images into mineable high-dimensional data, which can reflect underlying pathophysiology via quantitative analyses.



Larue RT, Defraene G, De Ruysscher D, Lambin P, van Elmpt W. Quantitative radiomics studies for tissue characterization: a review of technology and methodological procedures. Br J Radiol. 2017 Feb;90(1070):20160665. doi: 10.1259/bjr.20160665.

Radiomics

Radiomics feature



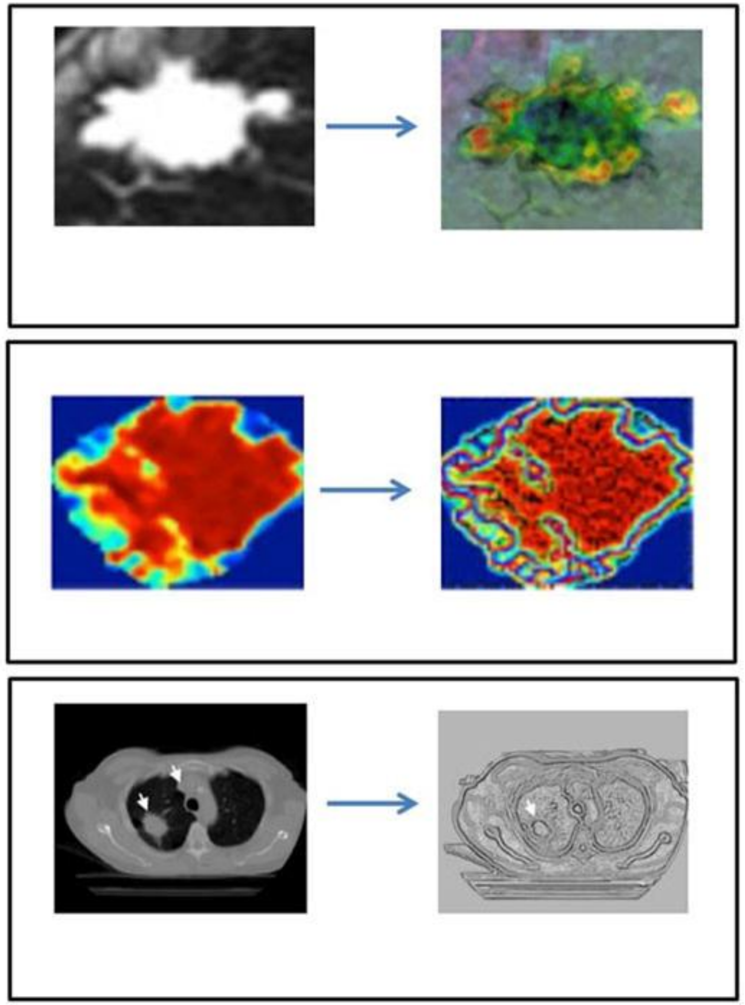
Yip SS, Aerts HJ. Applications and limitations of radiomics. Phys Med Biol. 2016 Jul 7;61(13):R150-66. doi: 10.1088/0031-9155/61/13/R150. Epub 2016 Jun 8. PMID: 27269645; PMCID: PMC4927328.

Filtering Approach

Statistical Kernel
(e.g. median filter)

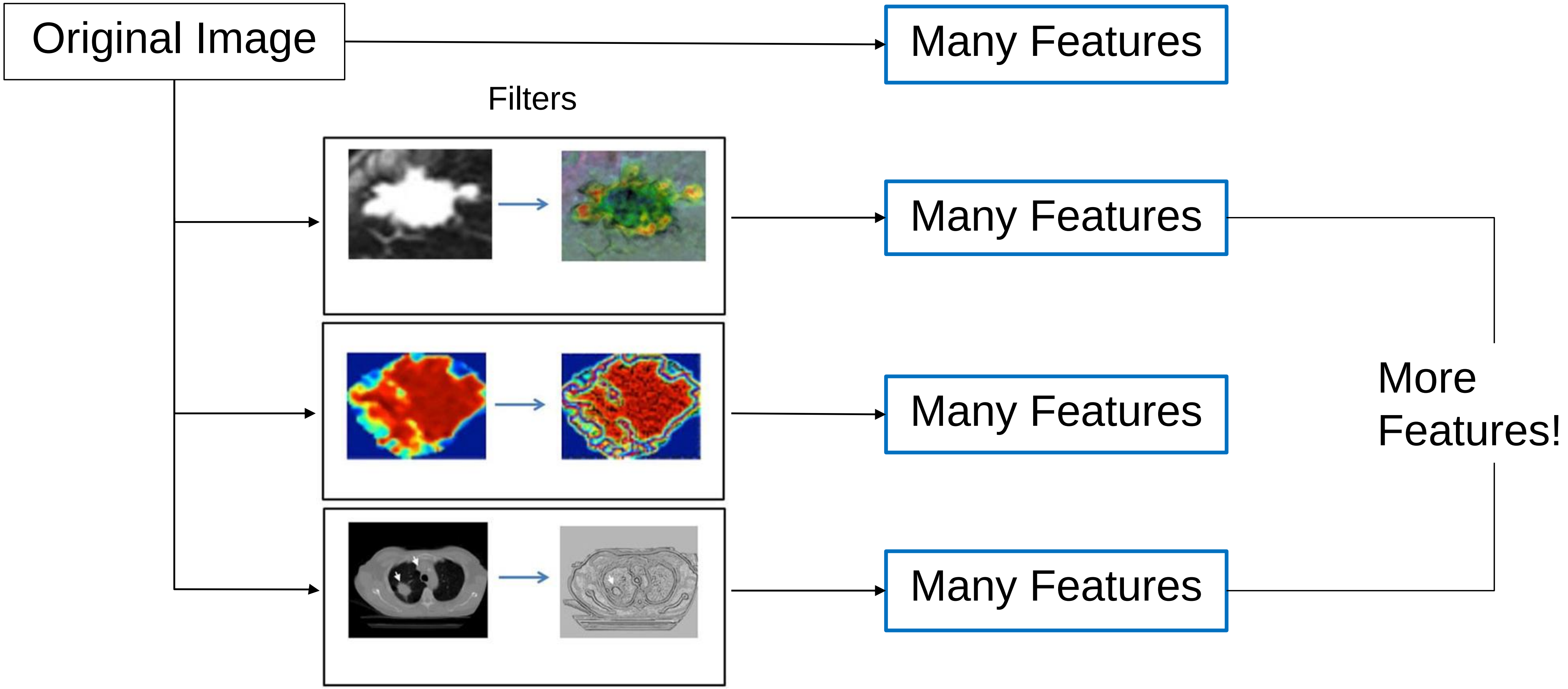
Edge Kernel
(e.g. Laplacian of Gaussian filter)

Special Kernel
(e.g. Fractal dimension filter)



Parekh V, Jacobs MA. Radiomics: a new application from established techniques. Expert Rev Precis Med Drug Dev. 2016;1(2):207-226. doi: 10.1080/23808993.2016.1164013.

Features x Filters



Parekh V, Jacobs MA. Radiomics: a new application from established techniques. Expert Rev Precis Med Drug Dev. 2016;1(2):207-226. doi: 10.1080/23808993.2016.1164013.

Radiomic Feature Extraction Preparation

Image Data

- MR, PET, CT
- Stage, Tracer, Contrast Phase

Segmentation

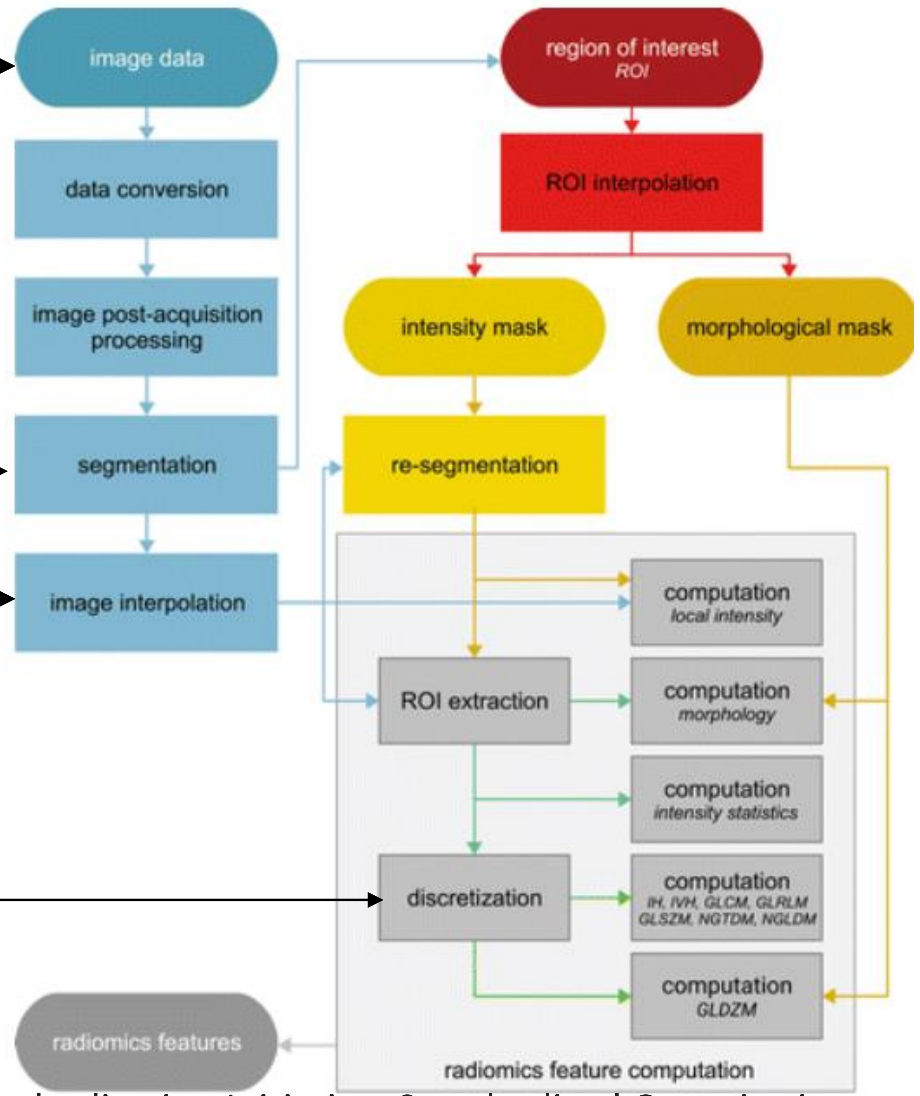
- Manual/Automatic Segmentation
- Radiologists, Segmentation Models

Interpolation (Resampling)

- Nearest Neighbor Interpolation
- Bilinear/Trilinear Interpolation

Discretization

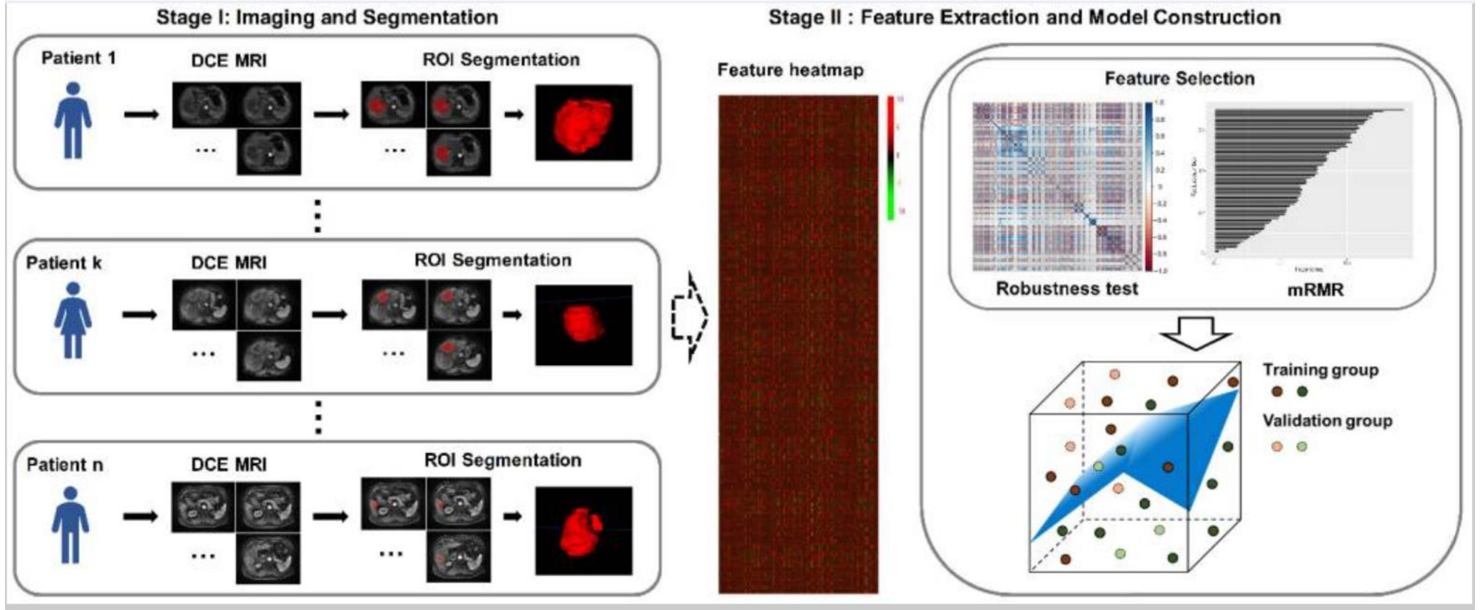
- Bin Width, Bin Number

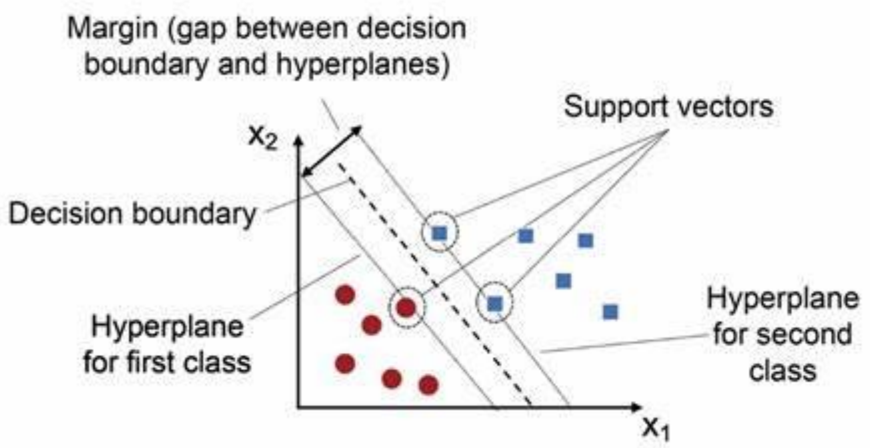


Zwanenburg A, Vallières M, Abdalah MA, et al. The Image Biomarker Standardization Initiative: Standardized Quantitative Radiomics for High-Throughput Image-based Phenotyping. Radiology. 2020 May;295(2):328-338.

Applications

- Tissue status evaluation (classification task, SVM)

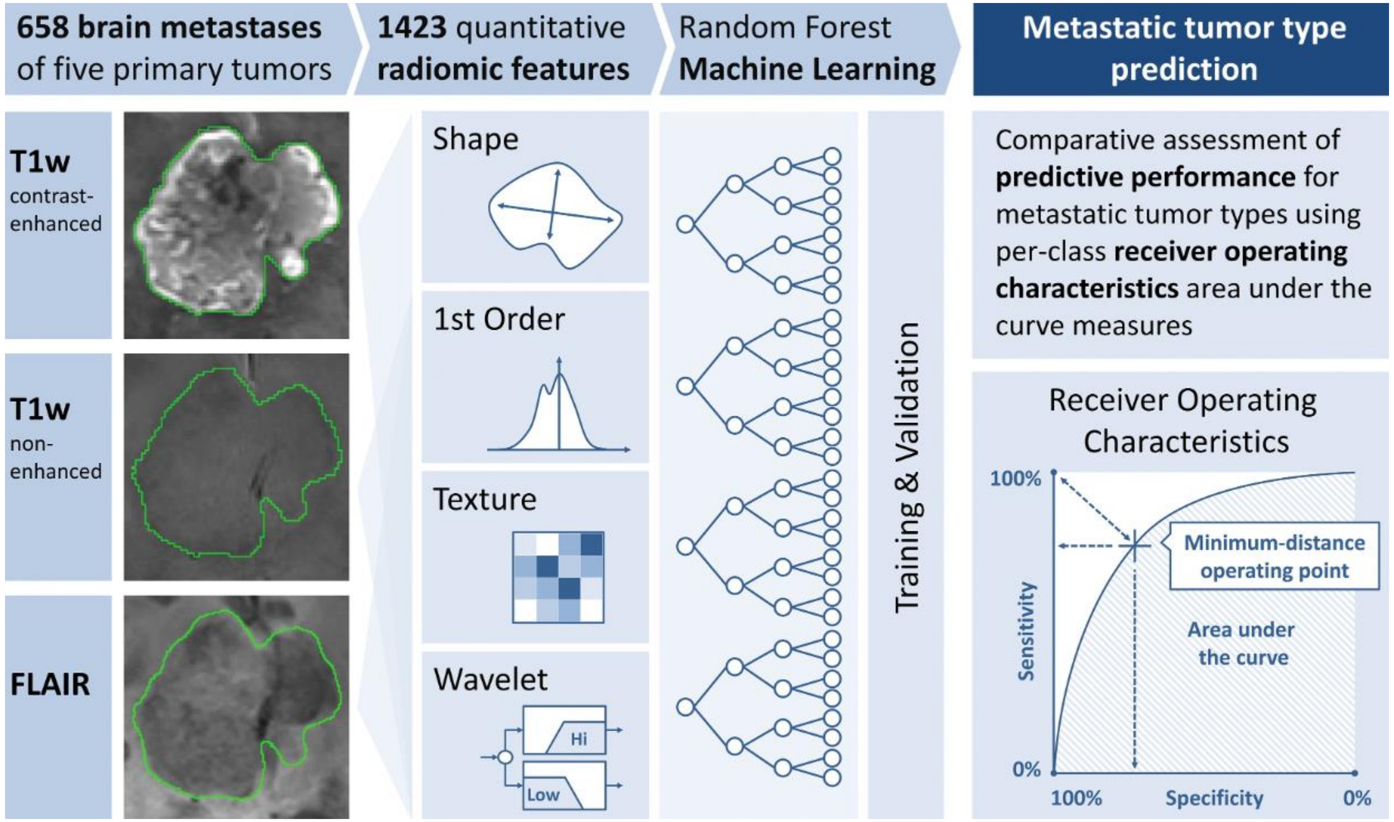


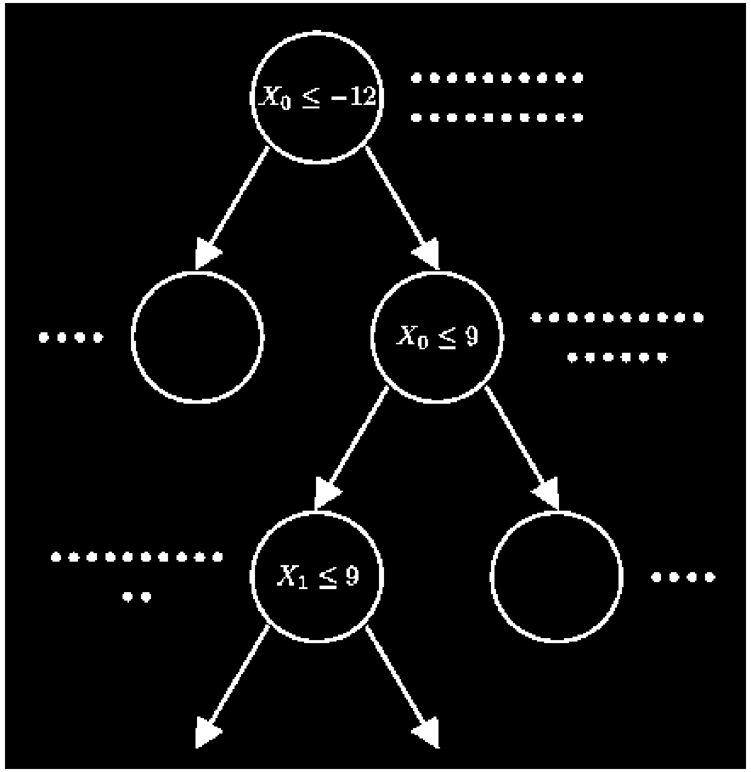


Xu L, Yang P, Liang W, et al. A radiomics approach based on support vector machine using MR images for preoperative lymph node status evaluation in intrahepatic cholangiocarcinoma[J]. Theranostics, 2019, 9(18): 5374.

Applications

- Predict tumor type (classification task, Random forest)

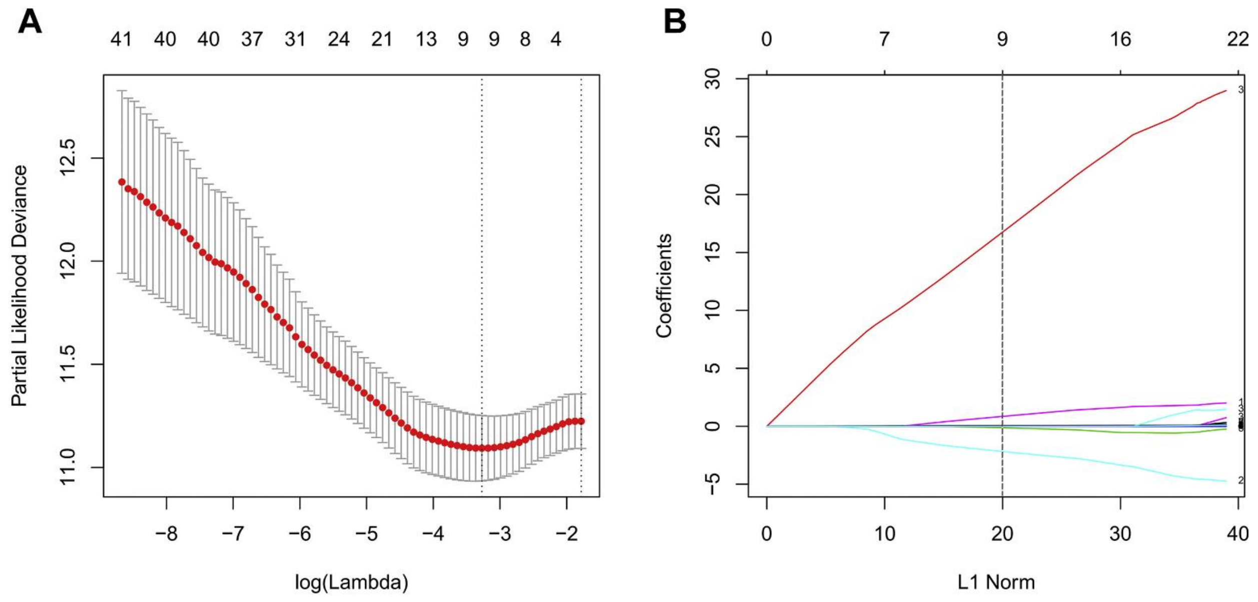


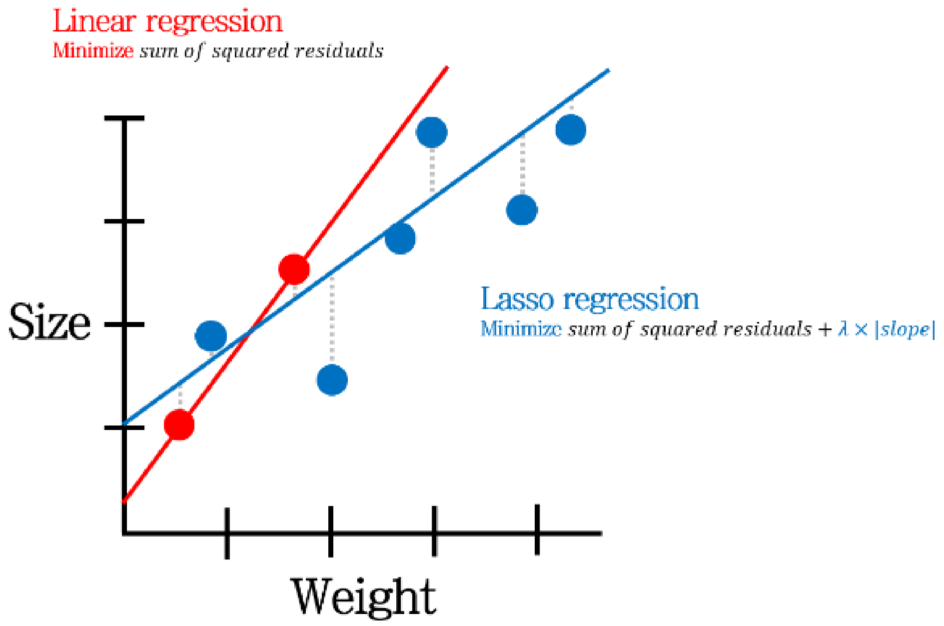


Kniep H C, Madesta F, Schneider T, et al. Radiomics of brain MRI: utility in prediction of metastatic tumor type[J]. Radiology, 2019, 290(2): 479-487.

Applications

- Predict progression-free survival (PFS) probability (regression task, LASSO)



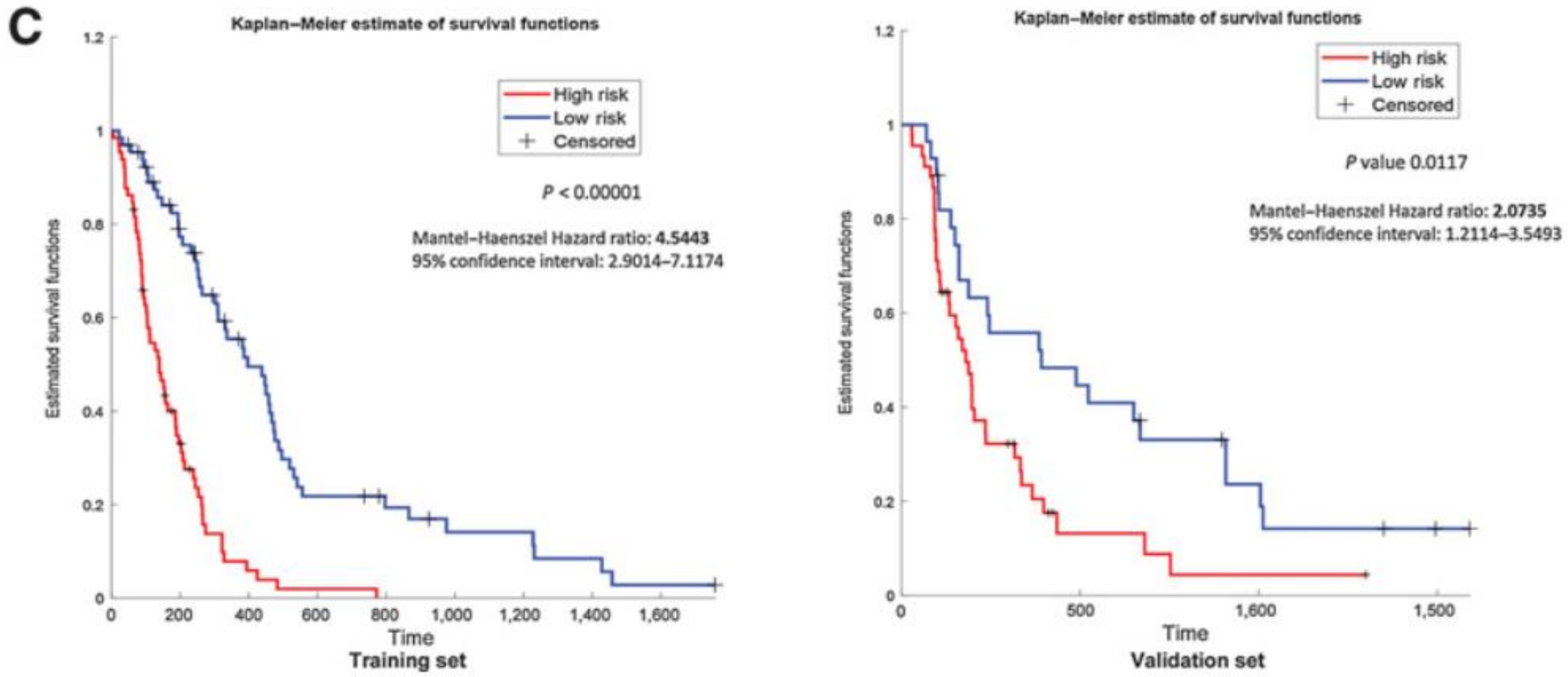


propose parameter "λ" as LASSO parameter for predict PFS

Liu X, Li Y, Qian Z, et al. A radiomic signature as a non-invasive predictor of progression-free survival in patients with lower-grade gliomas[J]. NeuroImage: Clinical, 2018, 20: 1070-1077.

Applications

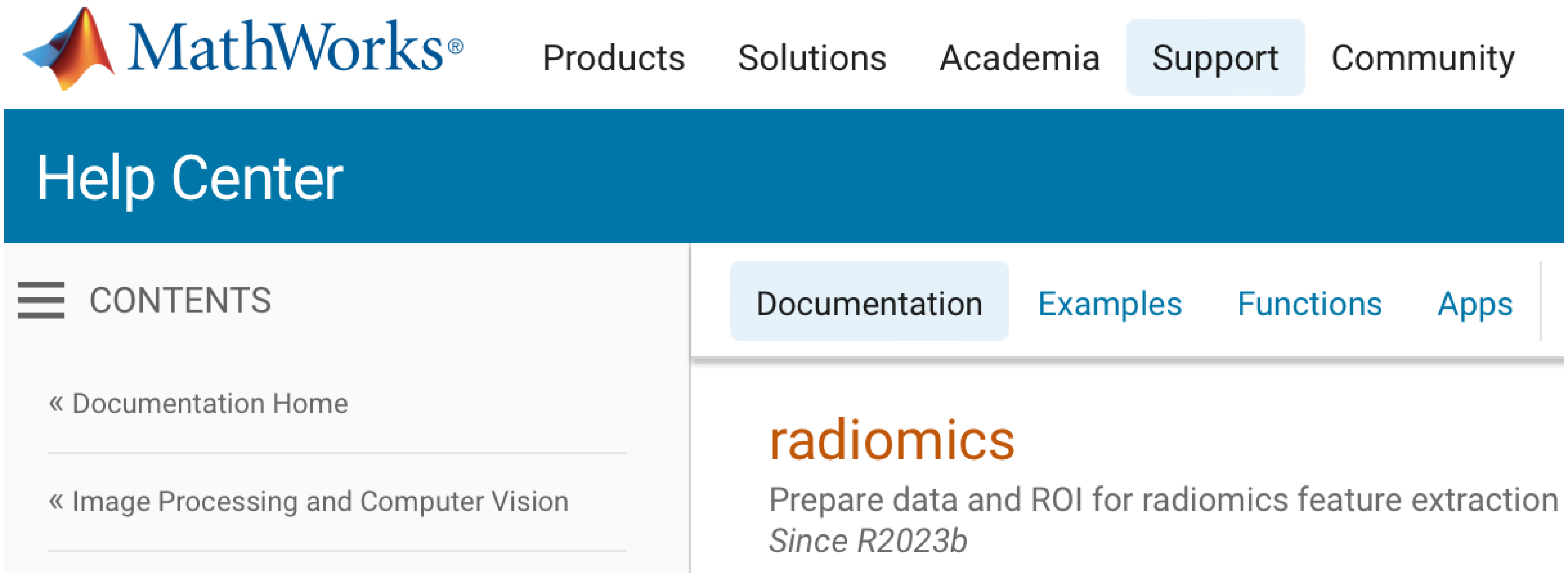
- Survival risk stratification (survival analysis)



Beig N, Bera K, Prasanna P, et al. Radiogenomic-based survival risk stratification of tumor habitat on Gd-T1w MRI is associated with biological processes in glioblastoma[J]. Clinical Cancer Research, 2020, 26(8): 1866-1876.

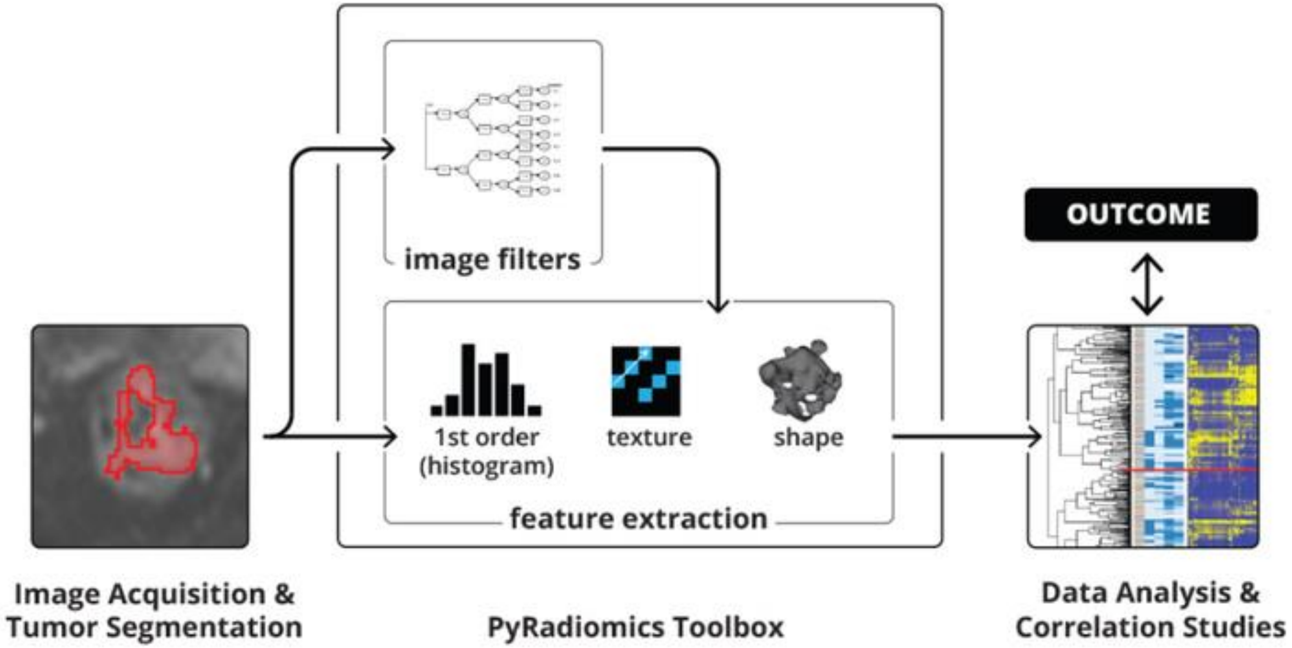
Tools

- Matlab



Tools

- pyradiomics

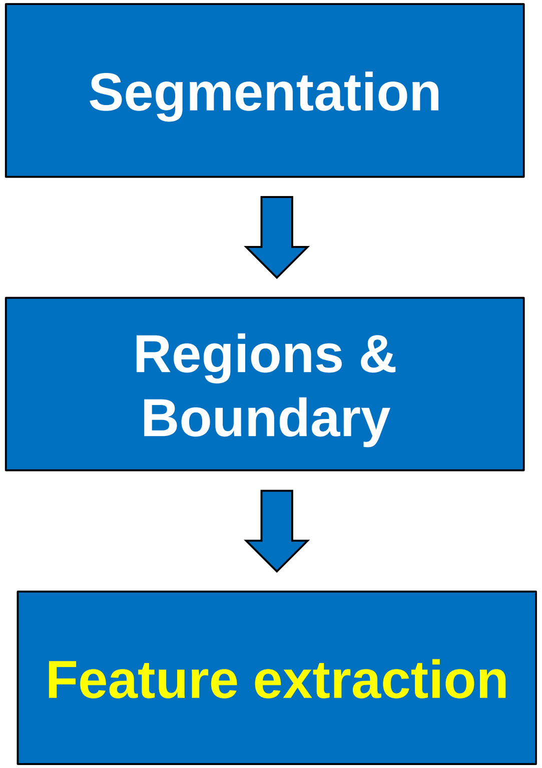


Limitations

- The **reproducibility** of radiomic studies is often poor
  - lack of standardization, insufficient reporting, or limited open source code and data
- Lack of proper **validation** and the subsequent risk of false-positive results
- **Interpretability** of the features
- **Retrospectively** collected data and thus have low level of evidence
- More

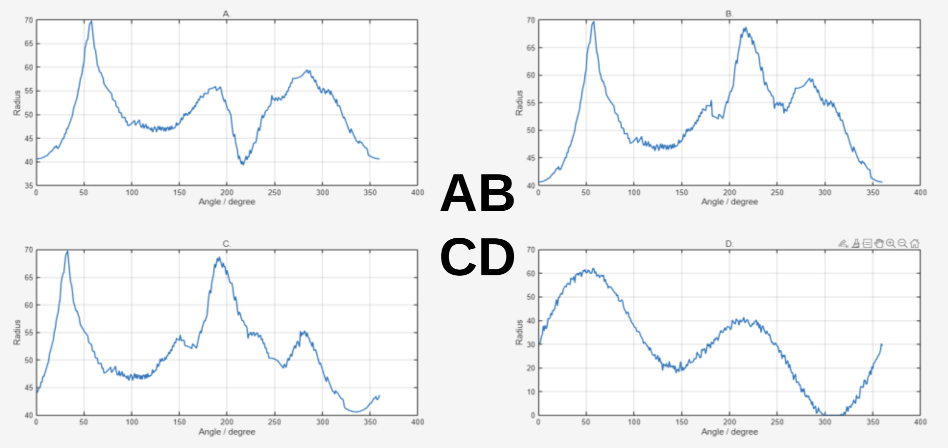
Outline

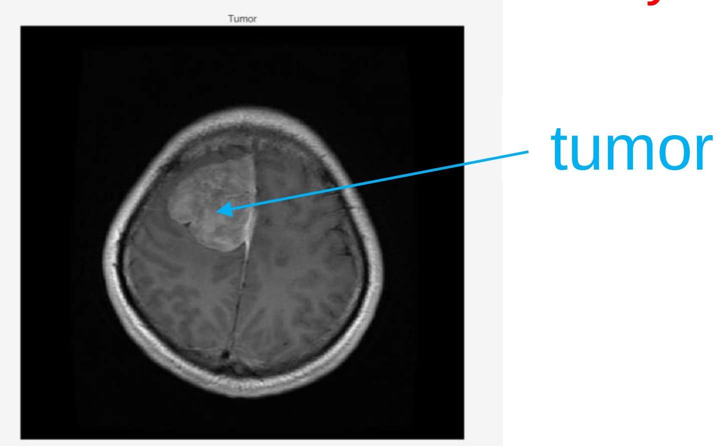
- Background
- Shape Feature
  - Boundary
  - Region
- Intensity Feature
- Texture Feature
  - GLCM
  - Fourier feature
- Radiomics

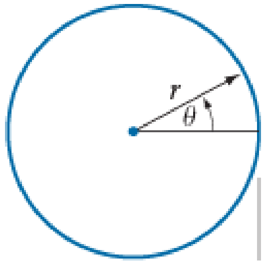


HW1- Signature and Fourier Descriptors

1.Which figure in is the true signature of the tumor? Choose the answer directly.

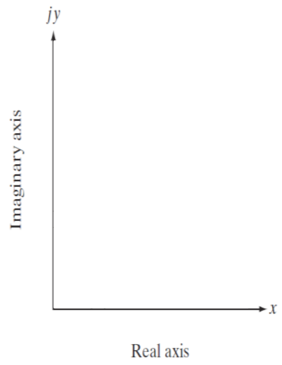






2.Plot the rectangular coordinate $f(x,y)$of the $r(\theta)$ (no need to derive $f(x,y)$)

$$r(\theta) = 2 - 2sin\theta + \frac{sin\theta\sqrt{|cos\theta|}}{sin\theta + 1.5}, \theta = \frac{2\pi n}{1280}, n = 0,1,2 \ldots 1279$$



3.Along the $f(x,y)$ boundary, get Fourier Descriptors. Set all $a(u) = 0$ except $a(0), a(1)$.

Plot the restored boundary and derive its equation. You can use matlab or python to help you calculate its coefficient. Keep two decimal places.

4.Replace $r(\theta)$ with $\rho(\theta) = \frac{1}{|\cos(\theta)|+|\sin(\theta)|}, \theta = \frac{2\pi n}{1280}, n = 0,1,2 \ldots 1279$, redo 2 and 3.

(***In your figures, student number as the watermark***).

HW2- First Order Features

1. Given the image I_0, carefully calculate its energy, range, μ_1, μ_2, μ_3, uniformity and entropy.

2. Assume that the image is a tiny temperature map on a world whose boundaries wrap around. The process of heat diffusion is analogous to applying circular convolution to an image.

Using circular padding: $I_{t+1} = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} *_{circular} I_t$

Which statement best describes this diffusion process from the perspective of images in practice?

A. It gently softens into the twilight, gradually losing its forms as its total brightness dissipates into the surrounding void.
B. As pixels endlessly intertwine in the cycle, the image awakens, stretching its dynamic range to forge sharper and deeper boundaries.
C. It does not perish in an instant; it slowly forgets its local forms, preserving its total brightness as a faint memory of the past, until all pixels drift toward the same quiet average — and the final whispers vanish into rounding and noise.
D. Like a drop of ink blooming in a calm pond, the image embraces absolute smoothness by continuously breathing new information into its canvas.

3.Fill in the table for I_∞ directly.

	energy	range	μ_1	μ_2	μ_3	uniformity	entropy
I_∞							

$$\begin{matrix} 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 2 & 0 & 3 & 1 \\ 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 3 & 1 \end{matrix}$$

I_0

HW3 - GLCM

- Find the co-occurrence matrix of a matrix pattern in the following cases.
- 1. The position operator Q is defined as "one pixel to the right"
- 2. The position operator Q is defined as "two pixels to the right"
- 3. For 1. and 2.'s GLCM, calculate contrast and homogeneity.

0	1	2	1	0
1	2	1	2	1
0	1	2	1	0
1	2	1	2	1
0	1	2	1	0

Matrix pattern

	0	1	2
0			
1			
2			

GLCM pattern